A production bias model of the Constant Rate Effect

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Context

- How can we make inferences about I-language on the basis of corpora?
- And can (diachronic) corpora tell us things that we wouldn't otherwise be able to discover?
- We need a **linking theory** that explains how grammatical knowledge is reflected in production data.
- The Constant Rate Hypothesis (Kroch 1989) is a seminal – but partial – theory of this kind.

Introduction

Aims of this talk:

- Outline some problems with the Constant Rate Hypothesis (CRH) as standardly described
- Provide a mechanism for Constant Rate Effects using a modified version of the learner in Yang (2000)
- Show that this model fits the data as well as or better than the traditional approach
- Show that interactions with time are *not* evidence for underlying grammatical disunity
- Illustrate an interesting consequence: when biases are in play, the grammar which can parse the most sentence types does not automatically win!

Part I

The Constant Rate Hypothesis and Constant Rate Effects

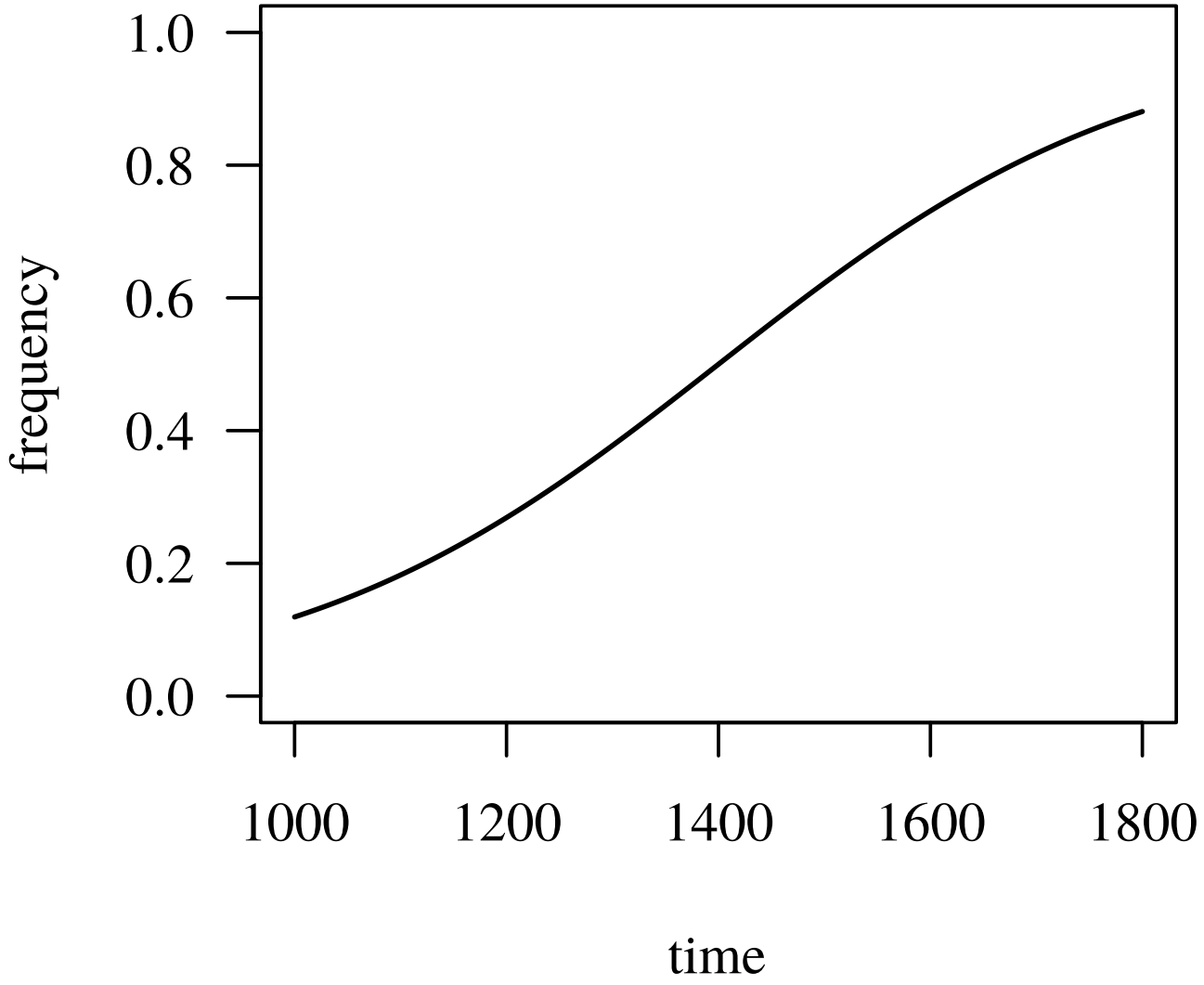
S-curves and the logistic equation

- Altmann et al. (1983), Kroch (1989): change is governed by the logistic equation.

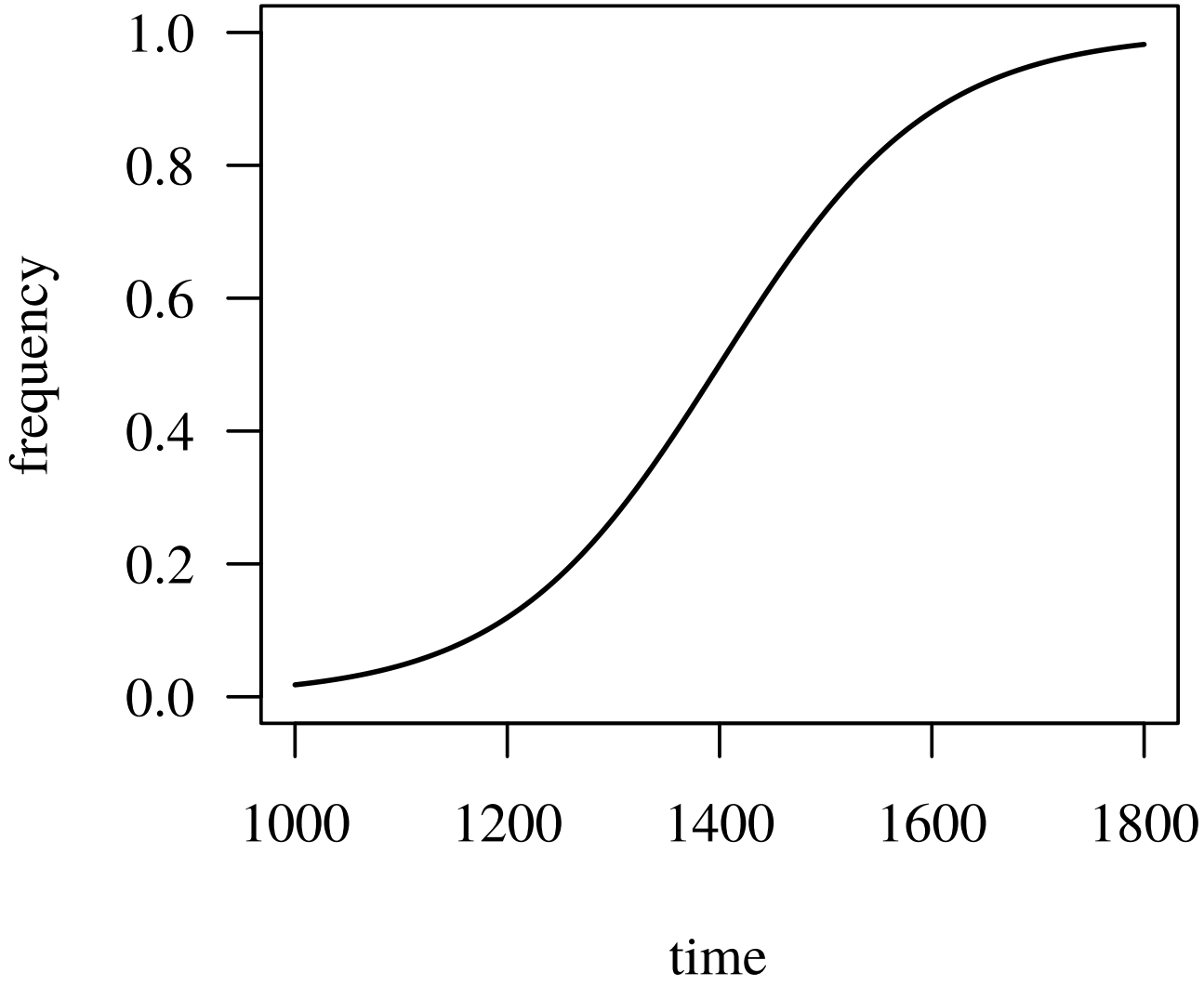
$$p(t) = \frac{1}{1 + e^{-s(t-k)}} \tag{1}$$

- The s parameter gives the overall steepness of the curve.
- The k parameter tells us where, in time, the curve is situated.

Varying the s parameter



Varying the s parameter



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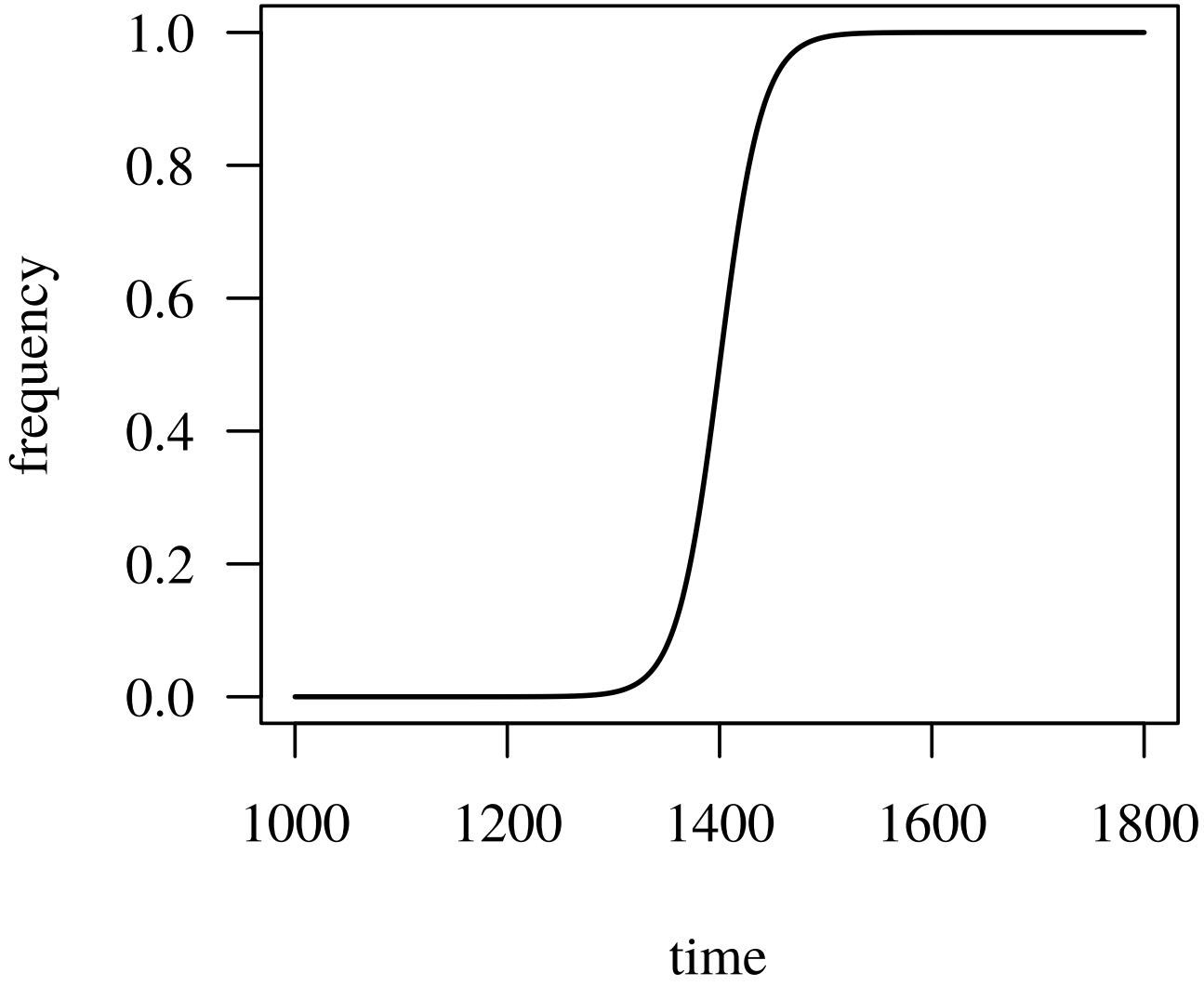
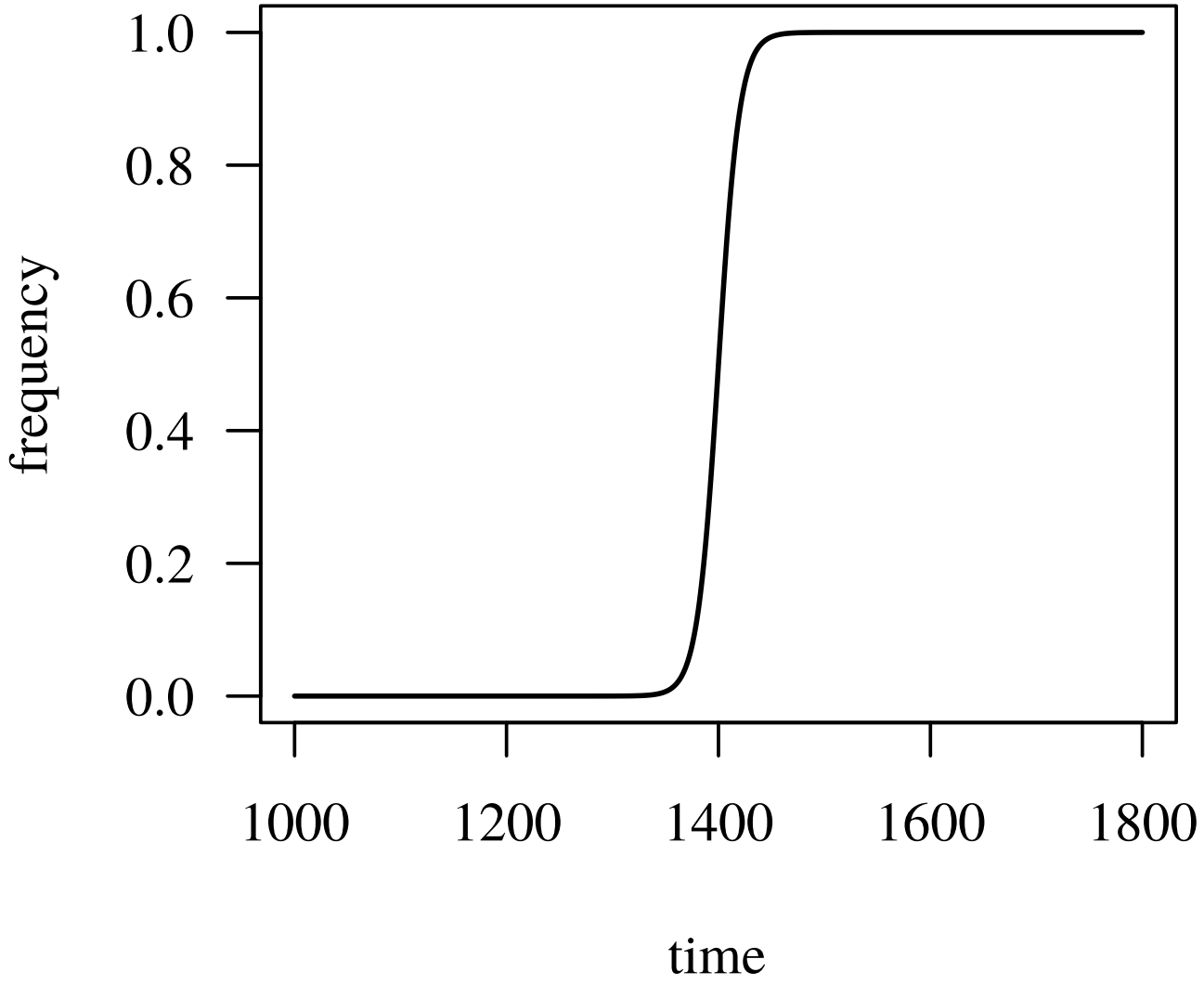
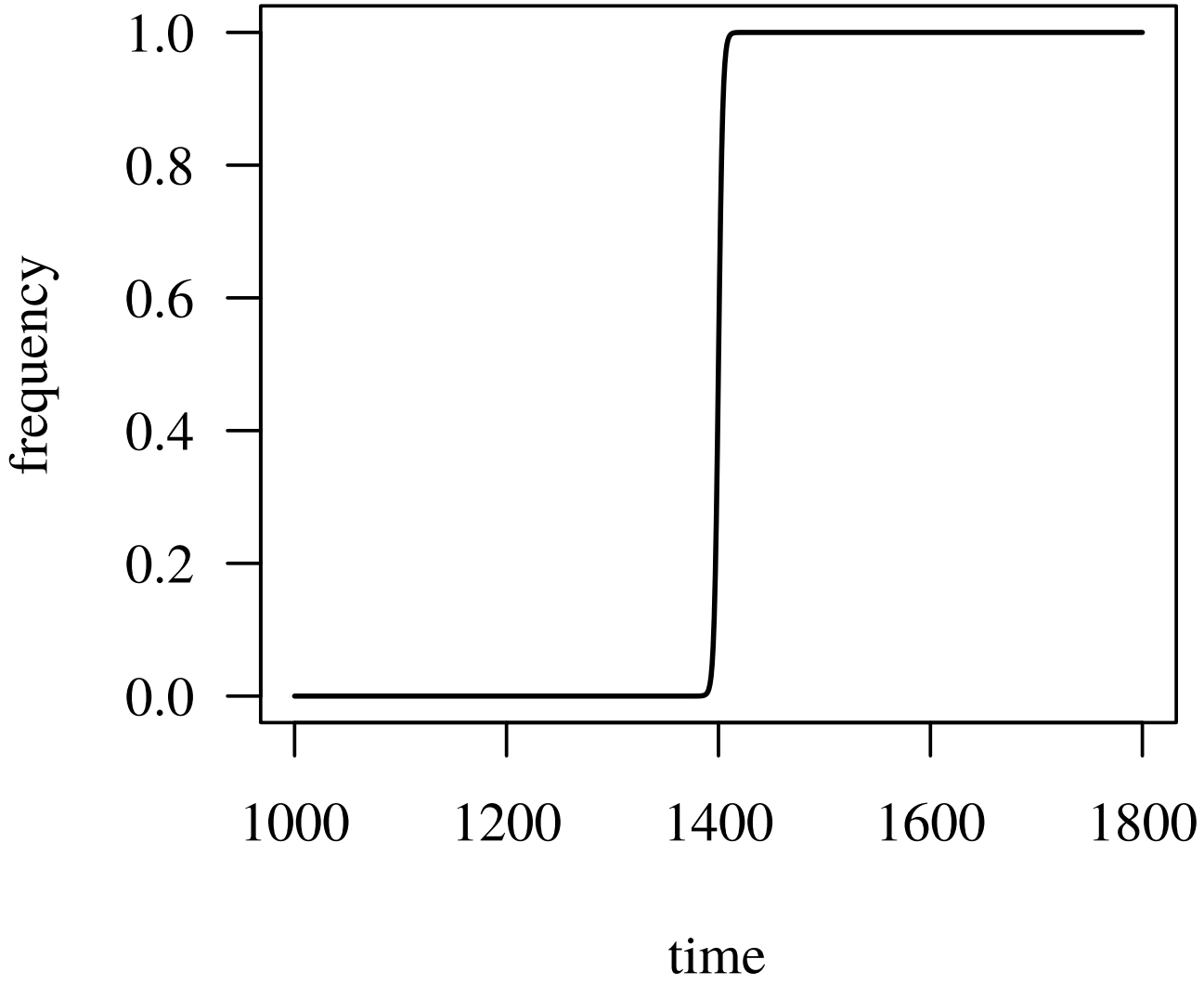


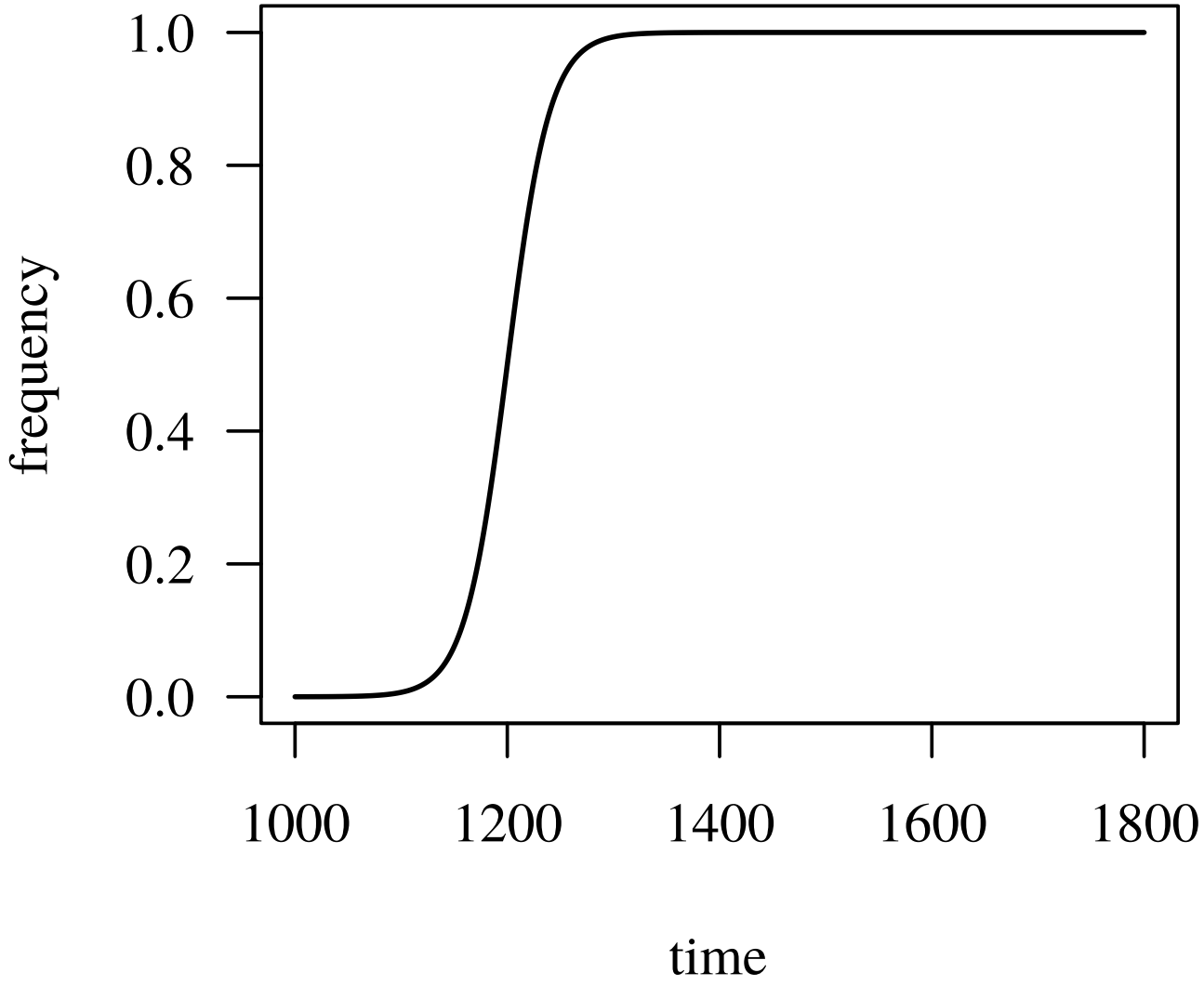
Varying the s parameter



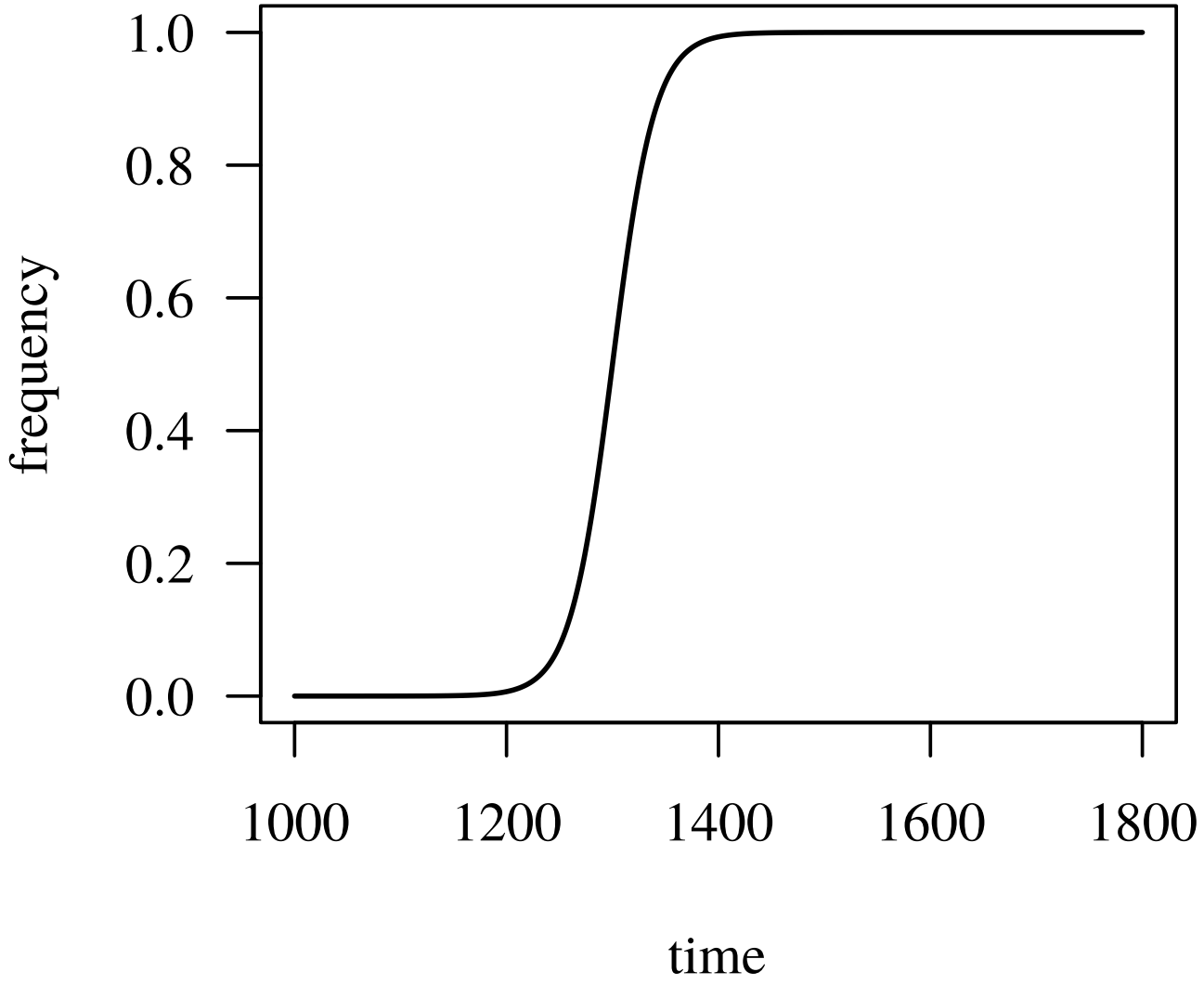
Varying the s parameter



Varying the k parameter



Varying the k parameter



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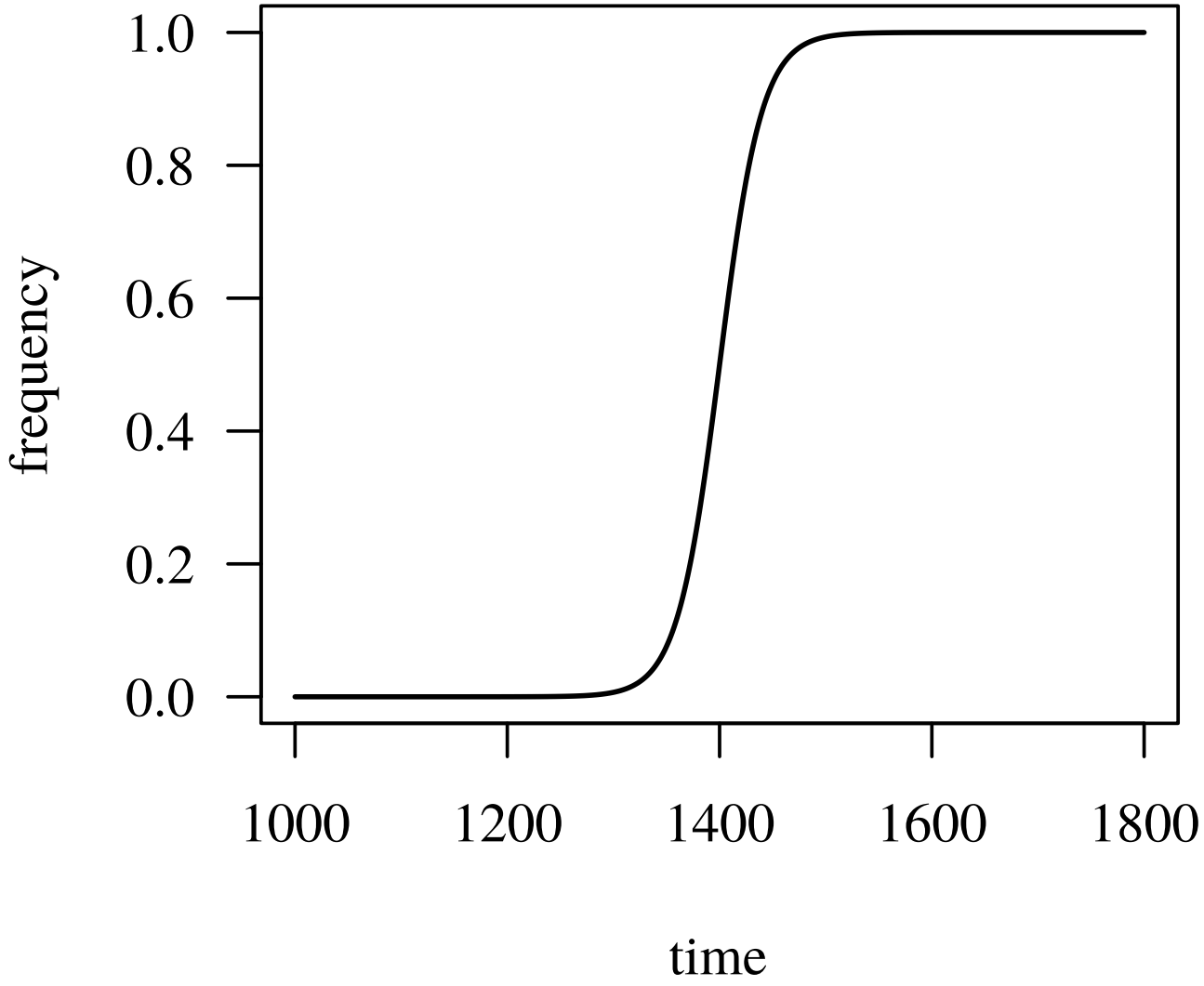
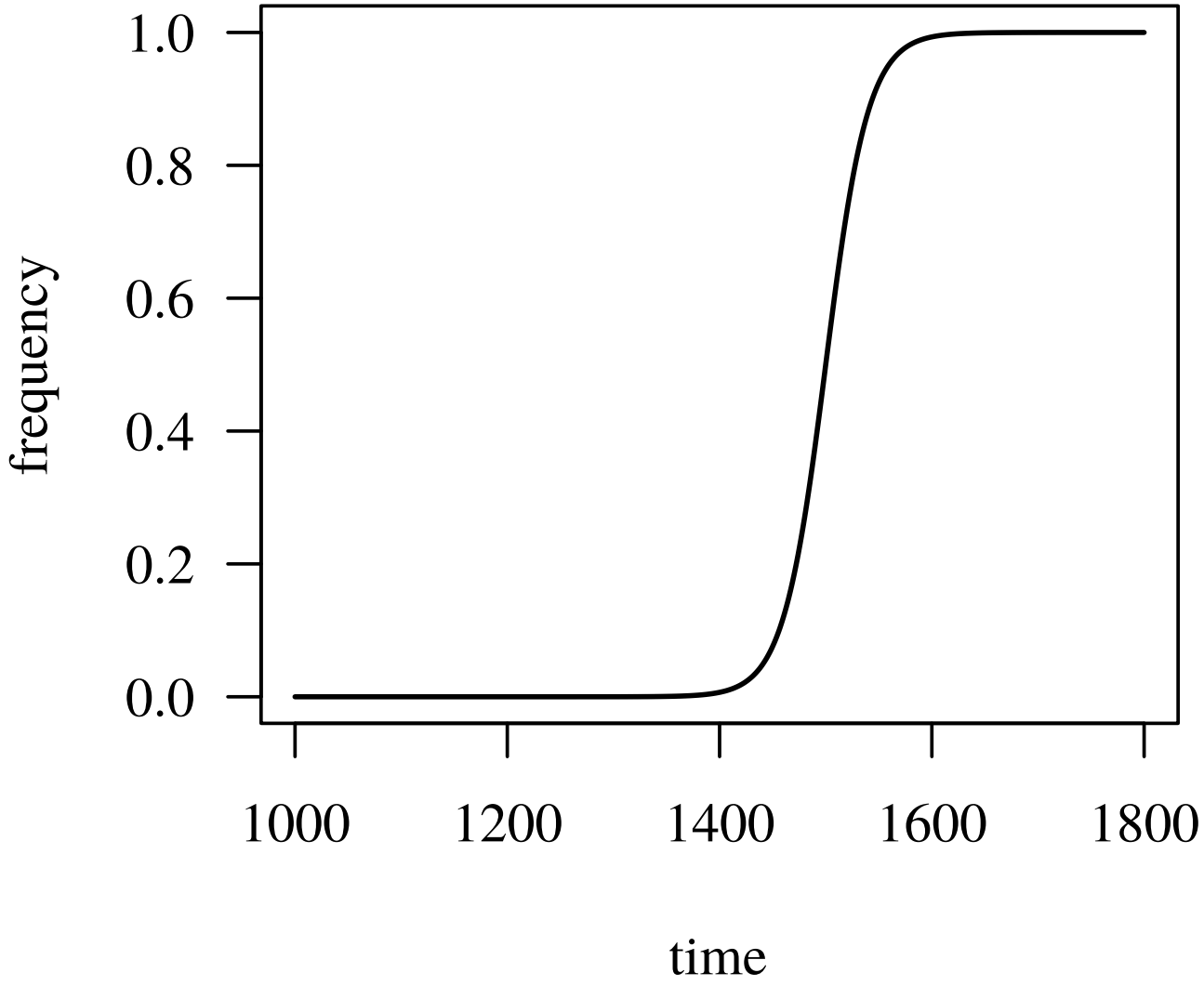
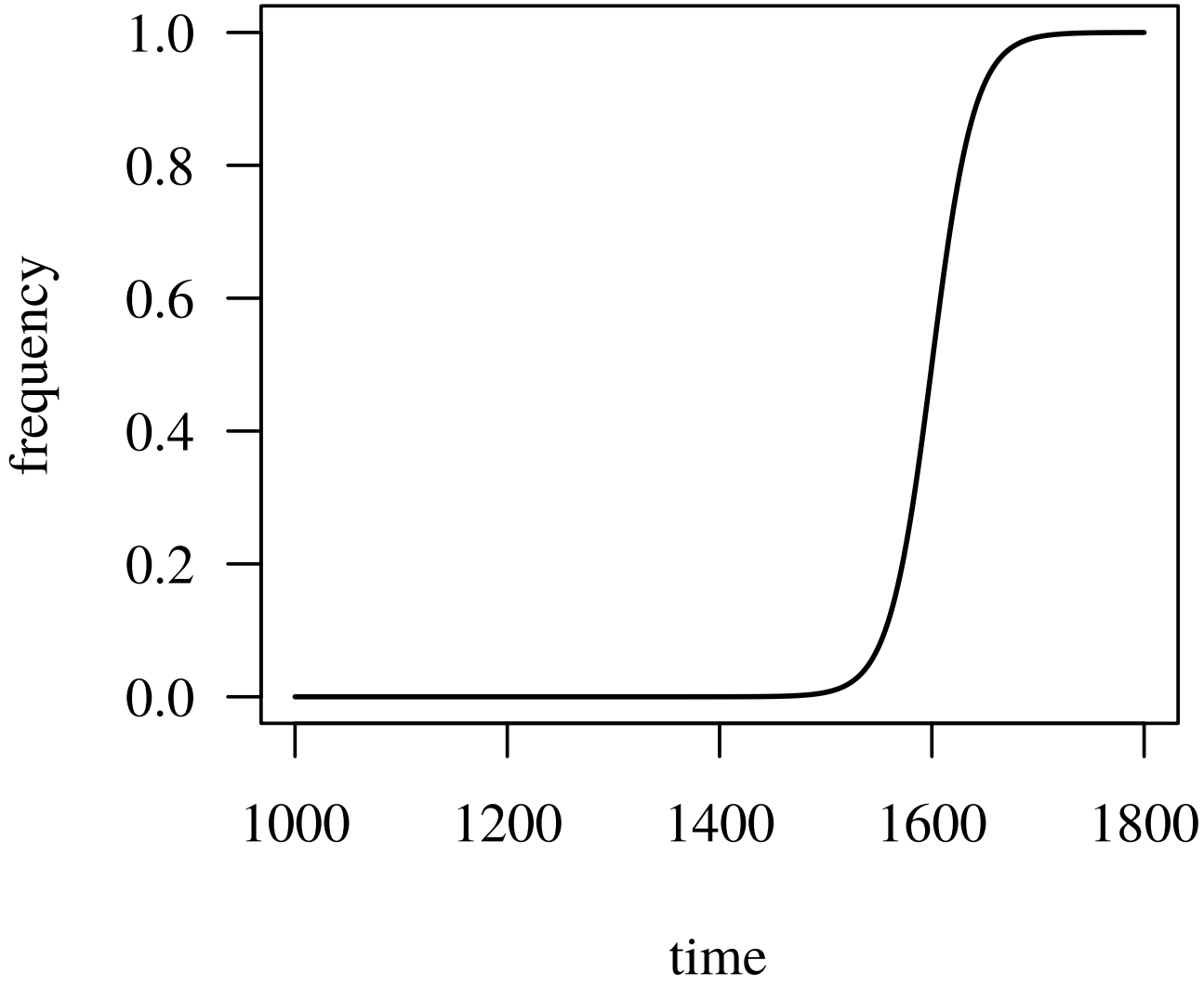


Varying the k parameter



Varying the k parameter

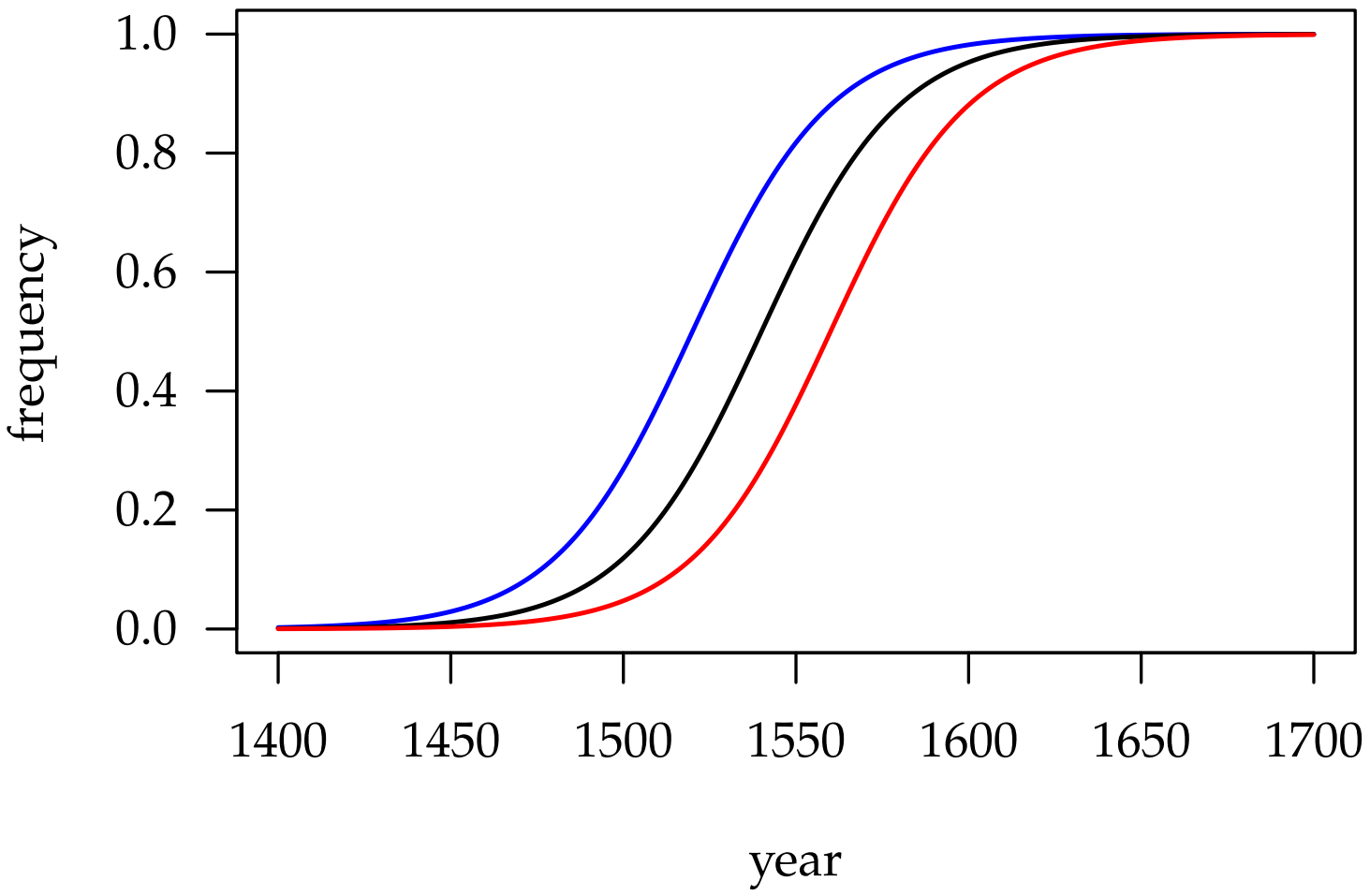


Constant Rate Hypothesis (CRH)

"[W]hen one grammatical option replaces another with which it is in competition across a set of linguistic contexts, the rate of replacement, properly measured, is the same in all of them." (Kroch, 1989, 200)

- In terms of the logistic equation: the s values are the same, but the k values may differ.
- Since Kroch (1989), many empirical studies have found this sort of situation (henceforth Constant Rate Effects, CREs).
- Interpretation (Kroch 1989): a single syntactic parameter governs all contexts. Some contexts may favour the new form, others disfavour it, but those (dis)favouring effects are **constant**.
- Relies on individual speakers having access to multiple, probabilistically-weighted **competing grammars**.

Different linguistic contexts

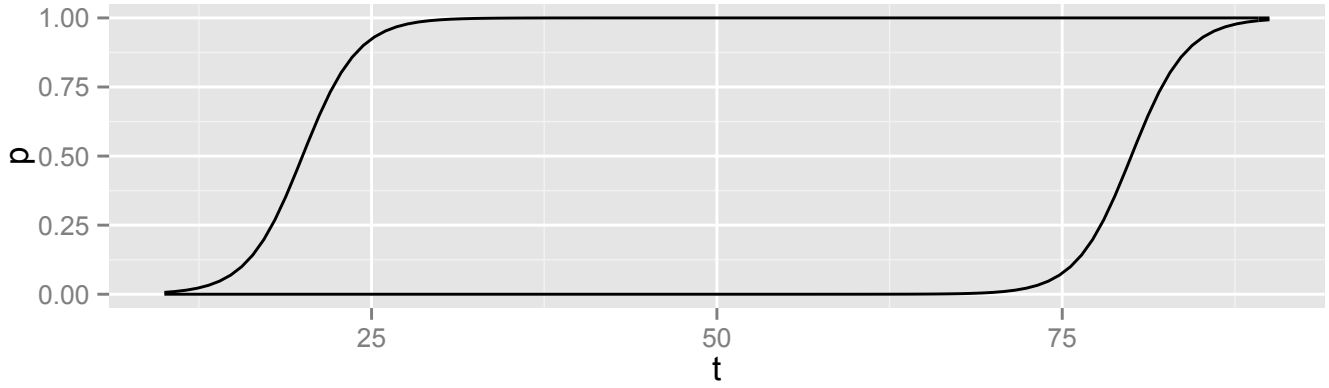


Central problem: no mechanism

- Kroch (1989, 204) adopts the logistic equation because of its use in biology, but admits that it does not follow from anything.
- Since then, Yang (2000) has derived the logistic equation from his model of syntactic acquisition and change (on which more soon).
- But no mechanism has so far been presented **to derive the CRH**: how exactly are the contexts tied to an underlying change?

Problem 1: time separation

- For a CRE, the s values must not differ significantly. But what about large differences in k?



- If the change only gets off the ground in one context 50 years after it has run to completion in another context (or 500, or 5,000,000), how plausible is it that we are dealing with reflexes of a single parametric change? (Think of the acquirer!)
- But this would still qualify as a CRE.

Problem 2: false positives?

CRE-like situations have been observed in situations in which the similar *s* values *cannot* be due to change in a single underlying parameter:

- "Contexts" are different geographical locations: Willis (2015)
- "Contexts" are different genders: Corley (2014)
- "Contexts" are different phenomena in different languages: Corley (2014)

Problem 3: the independence claim

Paolillo (2011):

- Standard way of testing for a CRE: chi-square test of independence on the s values of the regressions for the different contexts (Kroch, 1989; Santorini, 1993).
- If result is not statistically significant, then a CRE is assumed to hold (e.g. Kroch 1989, 225).
- BUT: non-significant results do not imply the validity of the null hypothesis! Possibility of type 2 error of unknown probability.

Part II

The model

Change

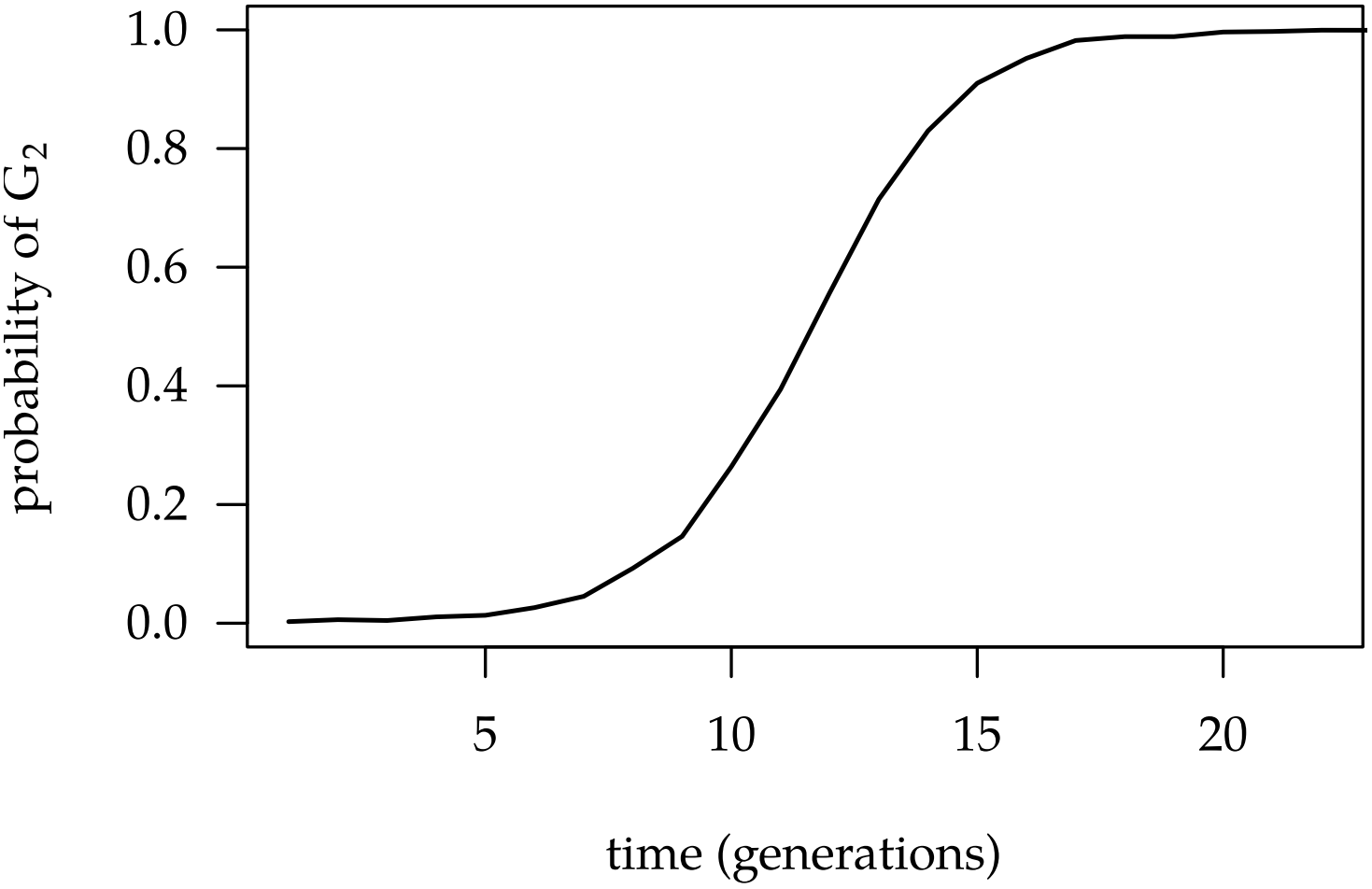
- Assume two grammatical hypotheses G_1 and G_2 compete in a population of speakers, so that G_1 is used with probability p and G_2 with probability q ($p + q = 1$)
- How do $p = p(t)$ and $q = q(t)$ change as a function of time t?
- Safe assumption: change is logistic, with either $p(t)$ growing and $q(t)$ decreasing or $q(t)$ growing and $p(t)$ decreasing as $t \to \infty$
- A plausible mechanism for change of this kind has been given by Yang (2000)
  - In this model, change is driven by the fact that different grammars parse different proportions of sentences the learner encounters in its environment

Change

- Generally, if change is logistic and generations are non-overlapping and discrete in time, a speaker of the $(t+1)$th generation has

$$p(t+1) = \left(1 + \rho \frac{1 - p(t)}{p(t)}\right)^{-1} \tag{2}$$

  (and similarly for $q(t+1)$), where ρ is a **growth rate** parameter
  - (2) can be solved to give the logistic equation (1) with $s = -\log(\rho)$ and $k = -\log(\rho)^{-1} \log(p(0)^{-1} - 1)$
- This is also true of Yang's (2000) model with ρ a parameter relating to the relative weak generative capacities of the competing grammars


1.0
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probability of G_2
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time (generations)

Change

- In fact, there is a strong prediction here regarding the outcome of grammatical competition
- The parameter ρ refers to the ratio of the weak generative capacities of the competing grammars G_1 and G_2 (called "advantages" in Yang, 2000)
  - if $\rho > 1$, then the proportion of sentences parsed by G_1 exclusively is larger than the proportion of sentences parsed by G_2 exclusively
  - if $\rho < 1$, vice versa
  - if $\rho = 1$, grammars "equally powerful"
- It can be shown (Yang, 2000):

Theorem (The Fundamental Theorem of Language Change)

Suppose learners are reliable (get a large number of input sentences and apply a reasonable learning rate). Then grammar G_2 overtakes grammar G_1 if, and only if, $\rho < 1$.

Production biases

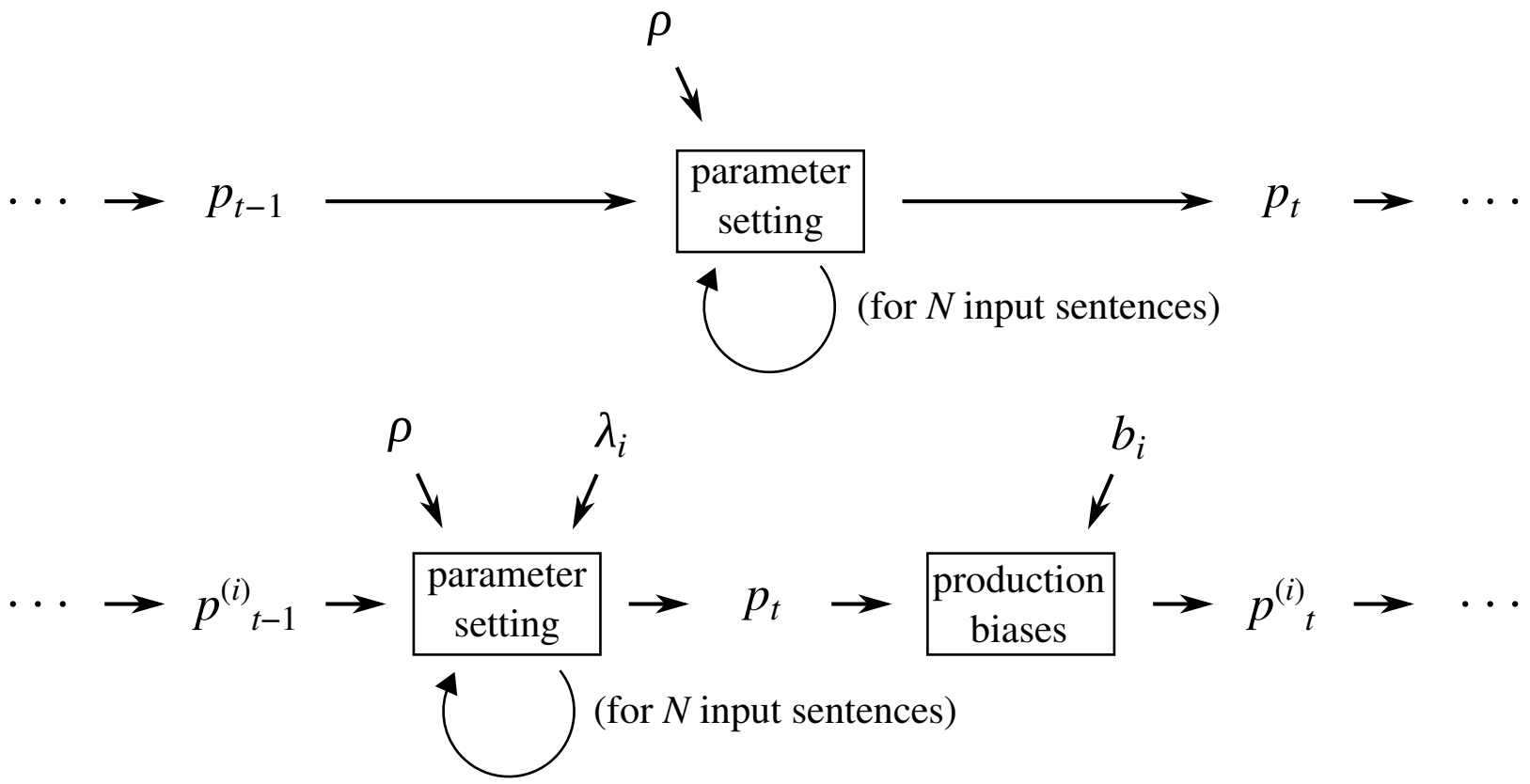
- What if there was an additional step in the process, where the learner's output is filtered by a **production bias**?
- I.e. learner emerges with a hypothesis (p, q) after learning, but in subsequent production filters this hypothesis according to some skewing mechanism that produces $(p - \epsilon, q + \epsilon)$ as the input for the following generation
- What if this biasing mechanism was sensitive to **linguistic contexts**?
  - E.g. different amount and/or direction of bias applied in questions and declaratives
- Intuitively, this should result in one of three outcomes:
  (1) Different contexts change at different rates
  (2) Different contexts begin to change at different times
  (3) Both (1) and (2)

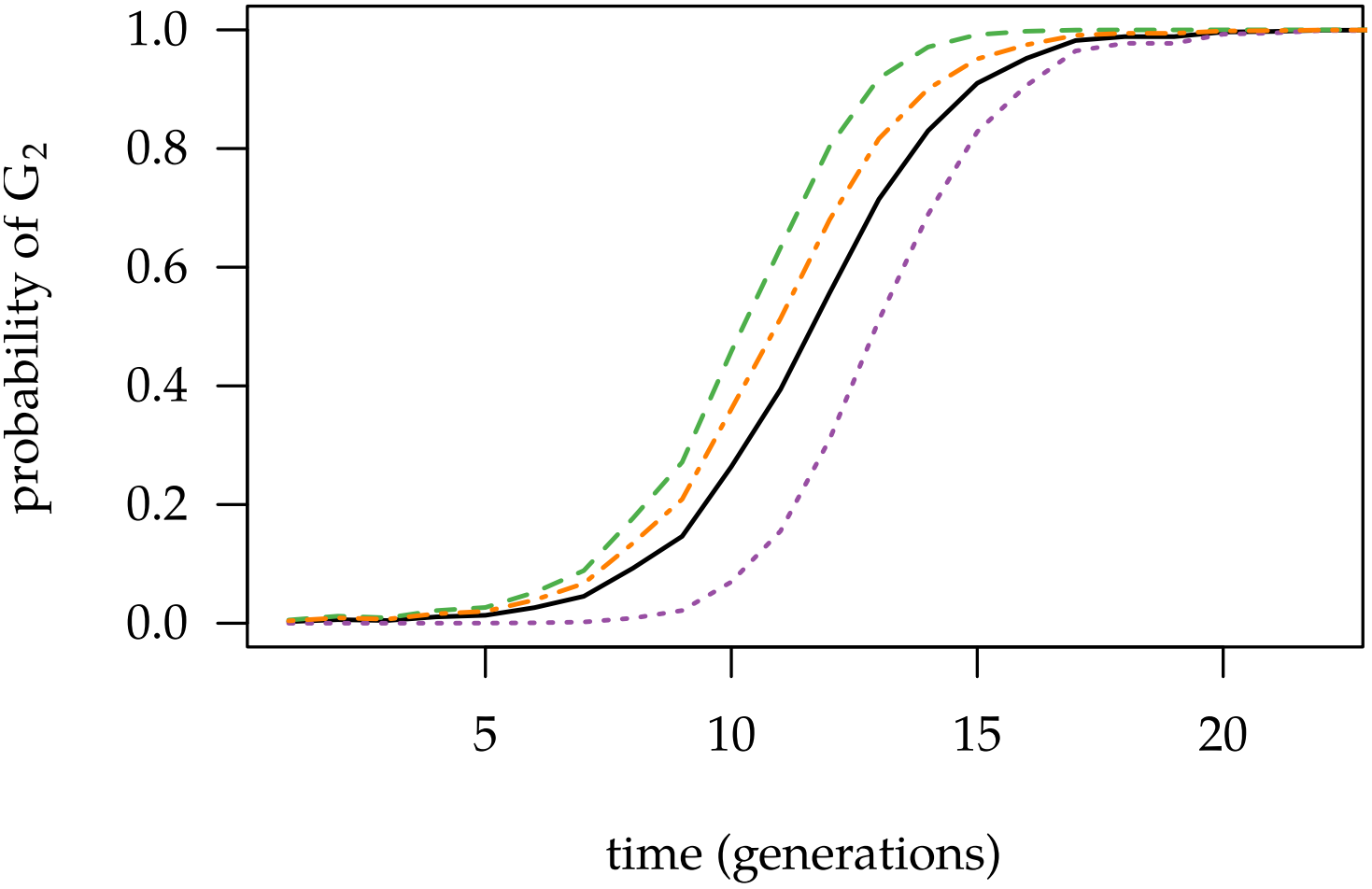
Production biases

- Assume K linguistic contexts $C_1, \ldots, C_K$
- Assume that with each context C_i is associated a constant (time-invariant) production bias b_i with the speaker's actual production in context C_i given by

$$\begin{cases} p_i = p - b_i p(1-p) \\ q_i = q + b_i q(1-q) \end{cases} \tag{3}$$

  with $-1 \leq b_i \leq 1$
- Intuition: degree of biasing for p_i has to depend on both p and $1-p$ (think of situations where $p = 0$ or $p = 1$)
- Notice:
  - if $0 < b_i \leq 1$ then context C_i favours G_2
  - if $-1 \leq b_i < 0$ context C_i favours G_1
  - if $b_i = 0$ there is no biasing in context C_i
  - if $p = 0$ or $q = 0$ there is no biasing in any context


ρ
$\cdots$
p_{t-1}
parameter setting
p_t
$\cdots$
(for N input sentences)
ρ
λ_i
b_i
$\cdots$
$p^{(i)}{}_{t-1}$
parameter setting
p_t
production biases
$p^{(i)}{}_t$
$\cdots$
(for N input sentences)


1.0
0.8
0.6
0.4
0.2
0.0
probability of G_2
5
10
15
20
time (generations)

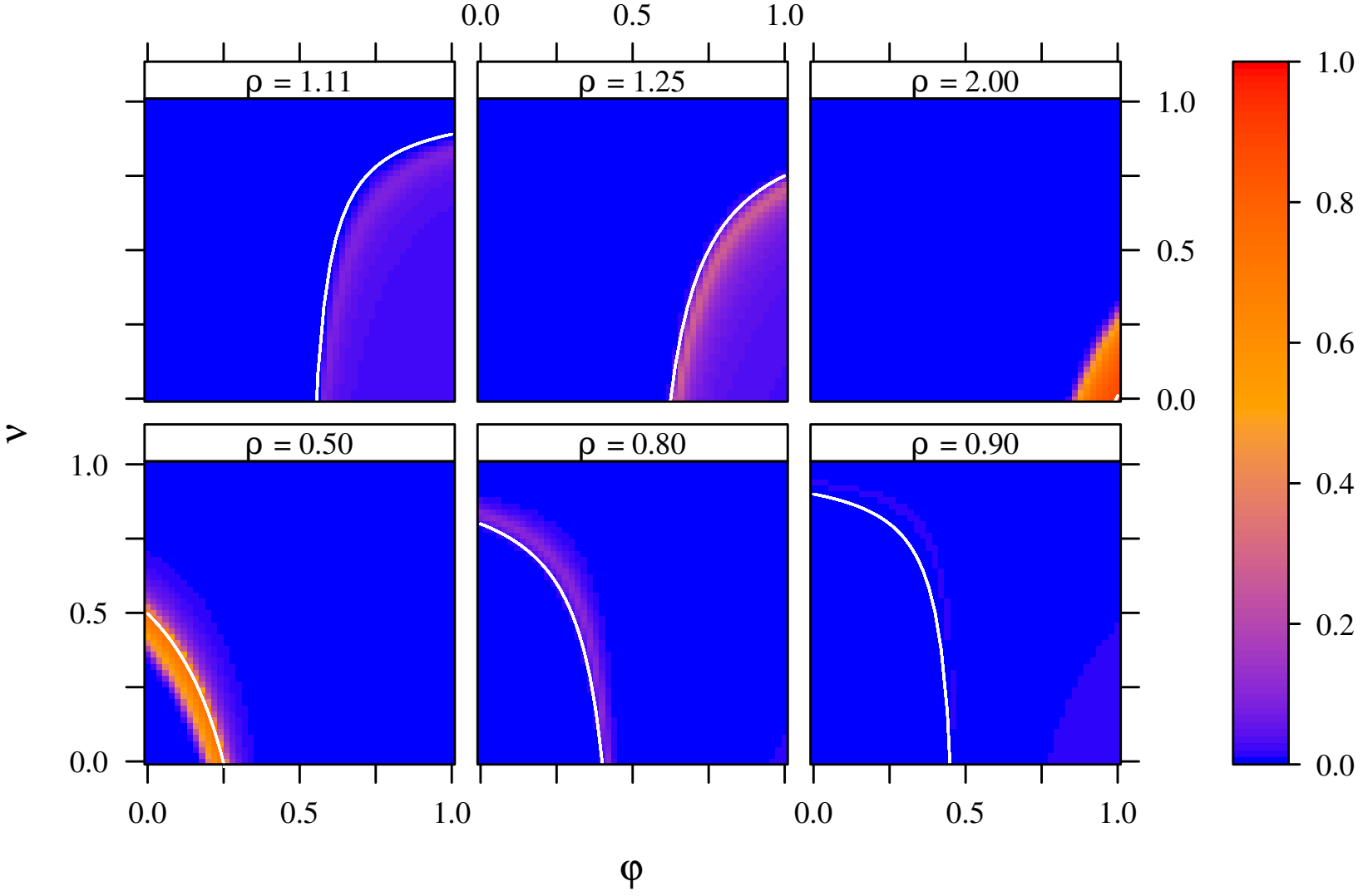
Production biases

- ICBS the logistic difference equation (2) is now replaced by the more complicated nonlinear equation

$$p(t+1) = \left(1 + \frac{1 - p(t)\sum_{i=1}^{K}\lambda_i b_i}{1 + (1 - p(t))\sum_{i=1}^{K}\lambda_i b_i}\rho\frac{1 - p(t)}{p(t)}\right)^{-1} \quad (4)$$

  with $q(t) = 1 - p(t)$ for all times t, where λ_i gives the proportion of sentences that fall in the ith context
- Crucial: there is a **feedback loop** between the underlying parametric change and the production biases, and the solution of (4) is not known
- However, we can replace (4) with the following logistic difference equation and establish (numerically) whether and how they differ:

$$\tilde{p}(t+1) = \left(1 + \frac{0.5\sum_{i=1}^{K}\lambda_i b_i}{1 + 0.5\sum_{i=1}^{K}\lambda_i b_i}\rho\frac{1 - \tilde{p}(t)}{\tilde{p}(t)}\right)^{-1} \quad (5)$$


ρ = 1.11
ρ = 1.25
ρ = 2.00
ρ = 0.50
ρ = 0.80
ρ = 0.90
ν
φ
0.0
0.5
1.0
0.2
0.4
0.6
0.8

The Time Separation Theorem

- Thus, the logistic approximation works very well for most combinations of model parameters
- This means we can derive each context curve $p_i(t)$ (resp. $q_i(t)$) from an underlying logistic curve $\tilde{p}(t)$ ($\tilde{q}(t)$) as follows:

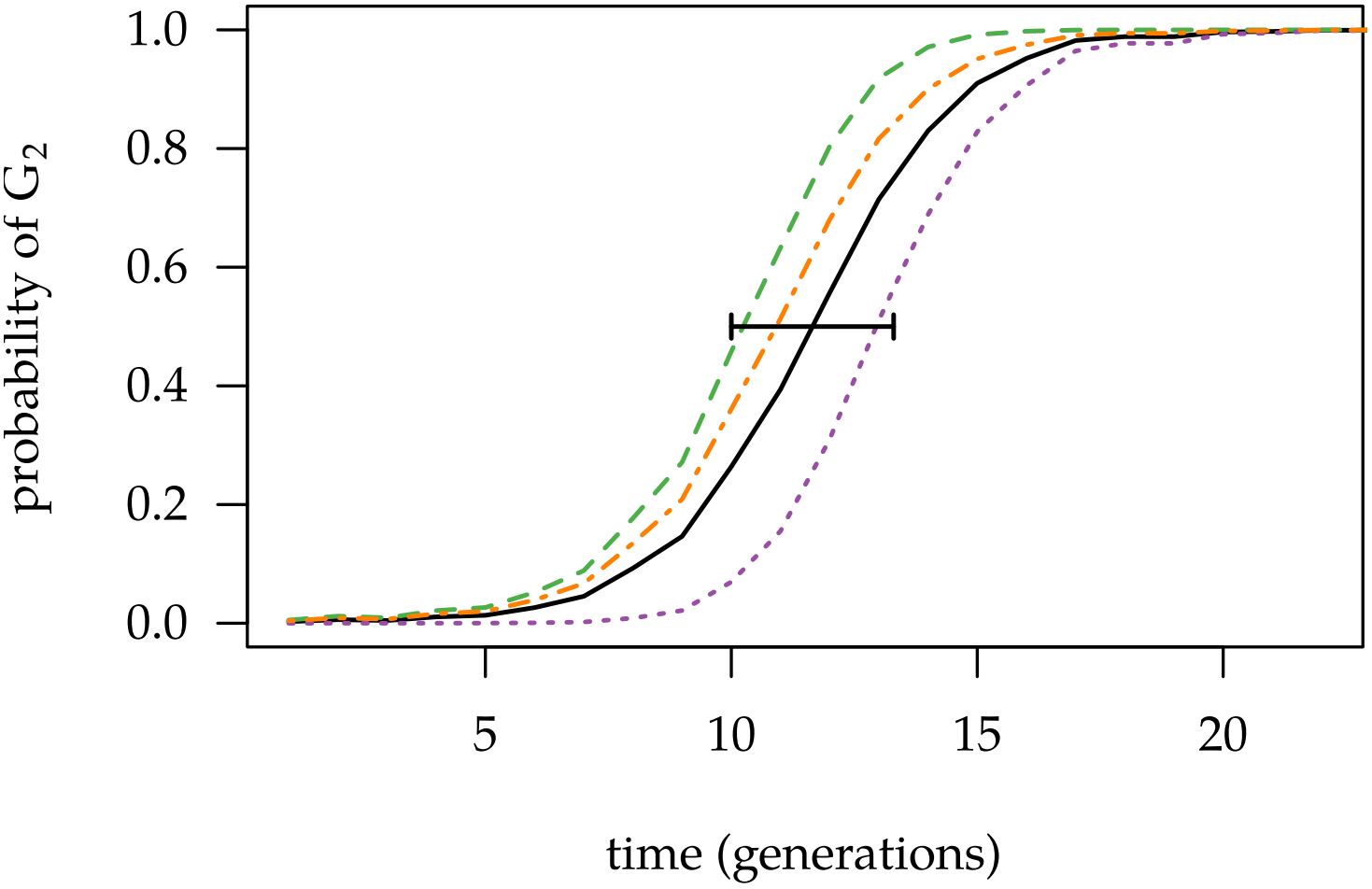
$$p_i(t) = \tilde{p}(t) + b_i \tilde{p}(t)(1 - \tilde{p}(t)) \tag{6}$$

  where $-1 \leq b_i \leq 1$.
- Importantly, this implies (Kauhanen & Walkden, 2015):

Theorem (The Time Separation Theorem)

For most values of the bias parameters b_i and λ_i, for any two contexts of an underlying change $p(t)$ proceeding at rate s, the maximal time separation at the tipping points is

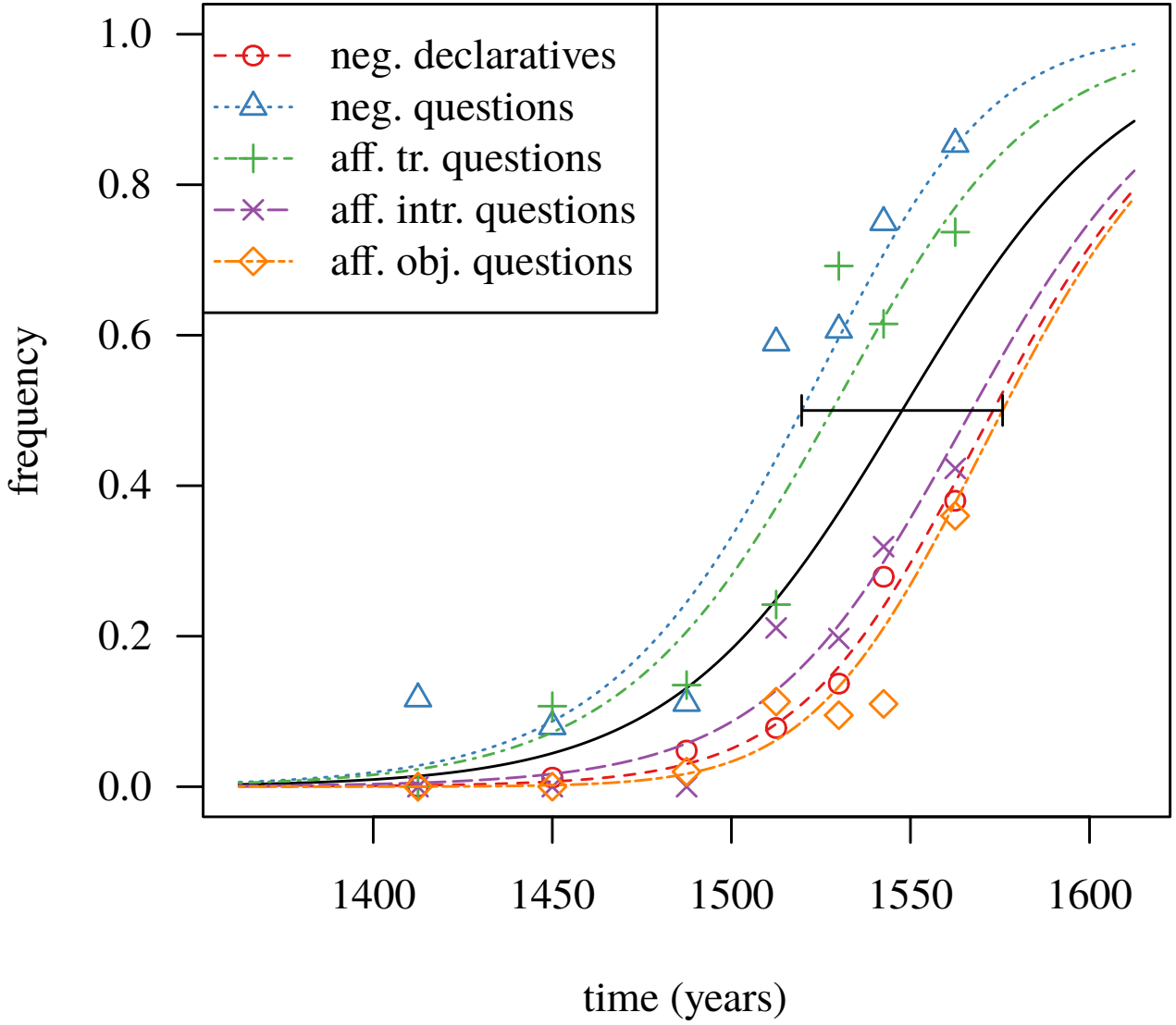
$$\Delta_{\max}(s) = \frac{2}{|s|} \log \left(\frac{1}{\sqrt{2} - 1} \right). \tag{7}$$


1.0
0.8
0.6
0.4
0.2
0.0
probability of G_2
5
10
15
20
time (generations)

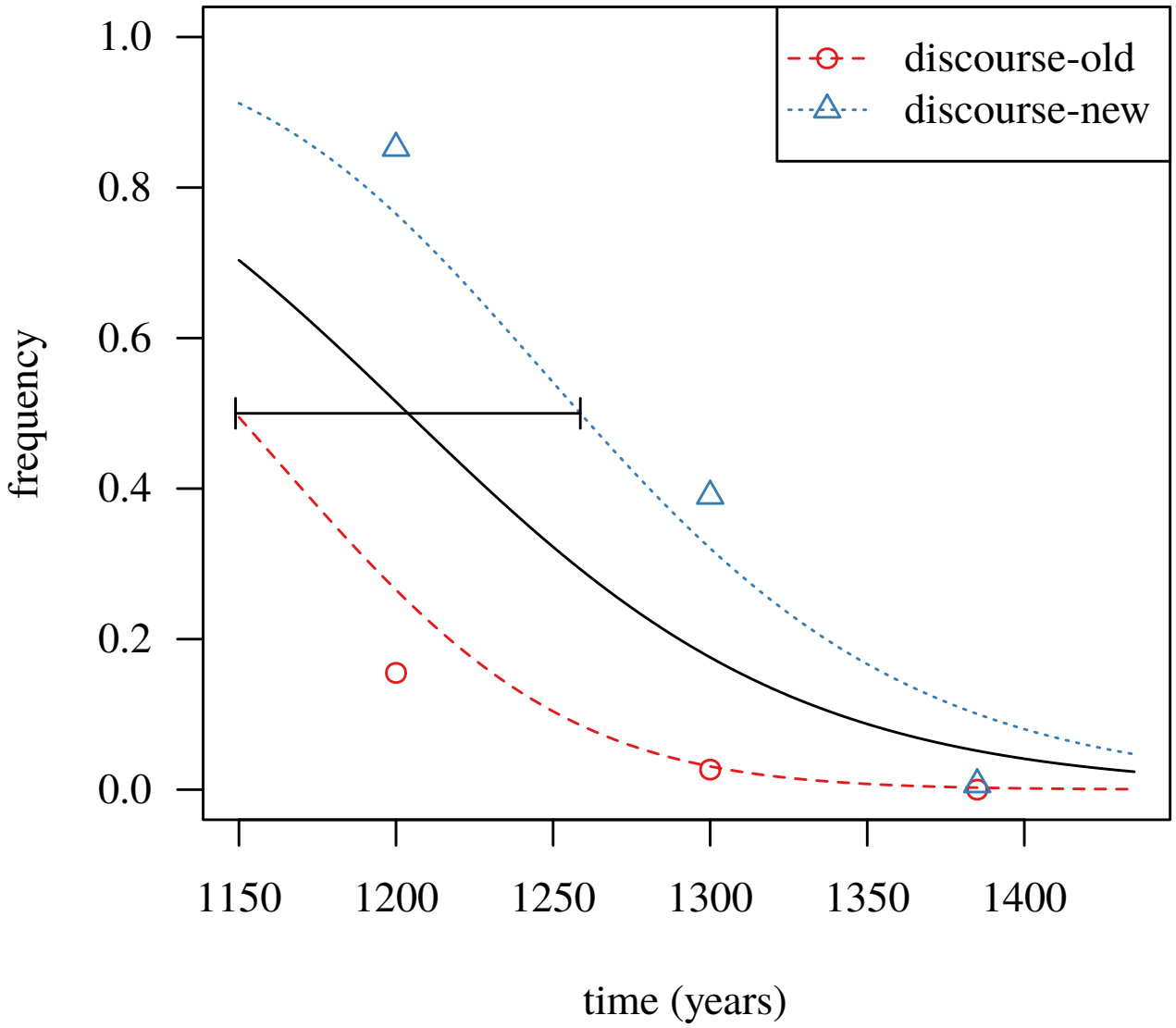
Part III

Evaluating the model: empirical fit

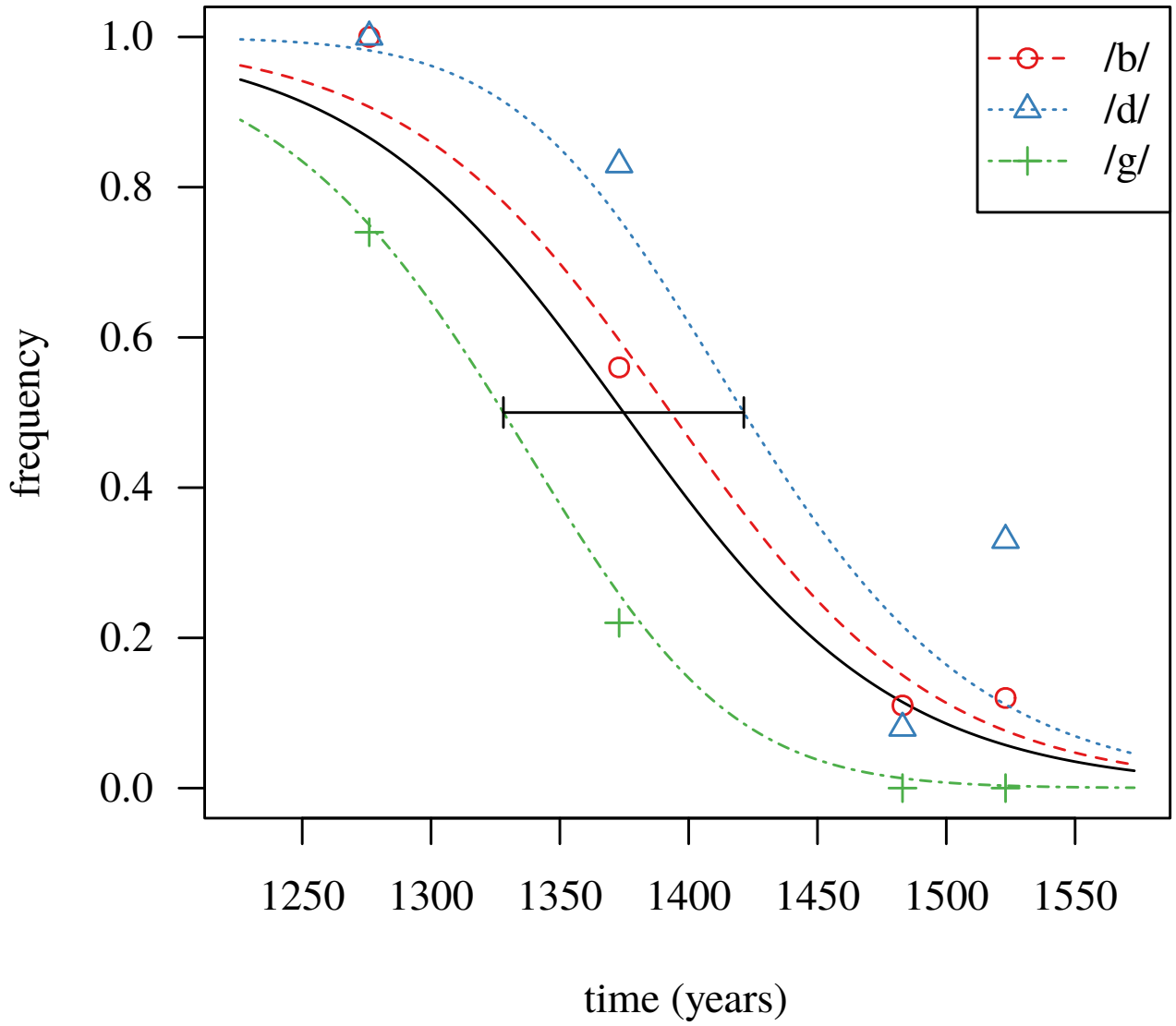
do-support in English (Kroch, 1989)



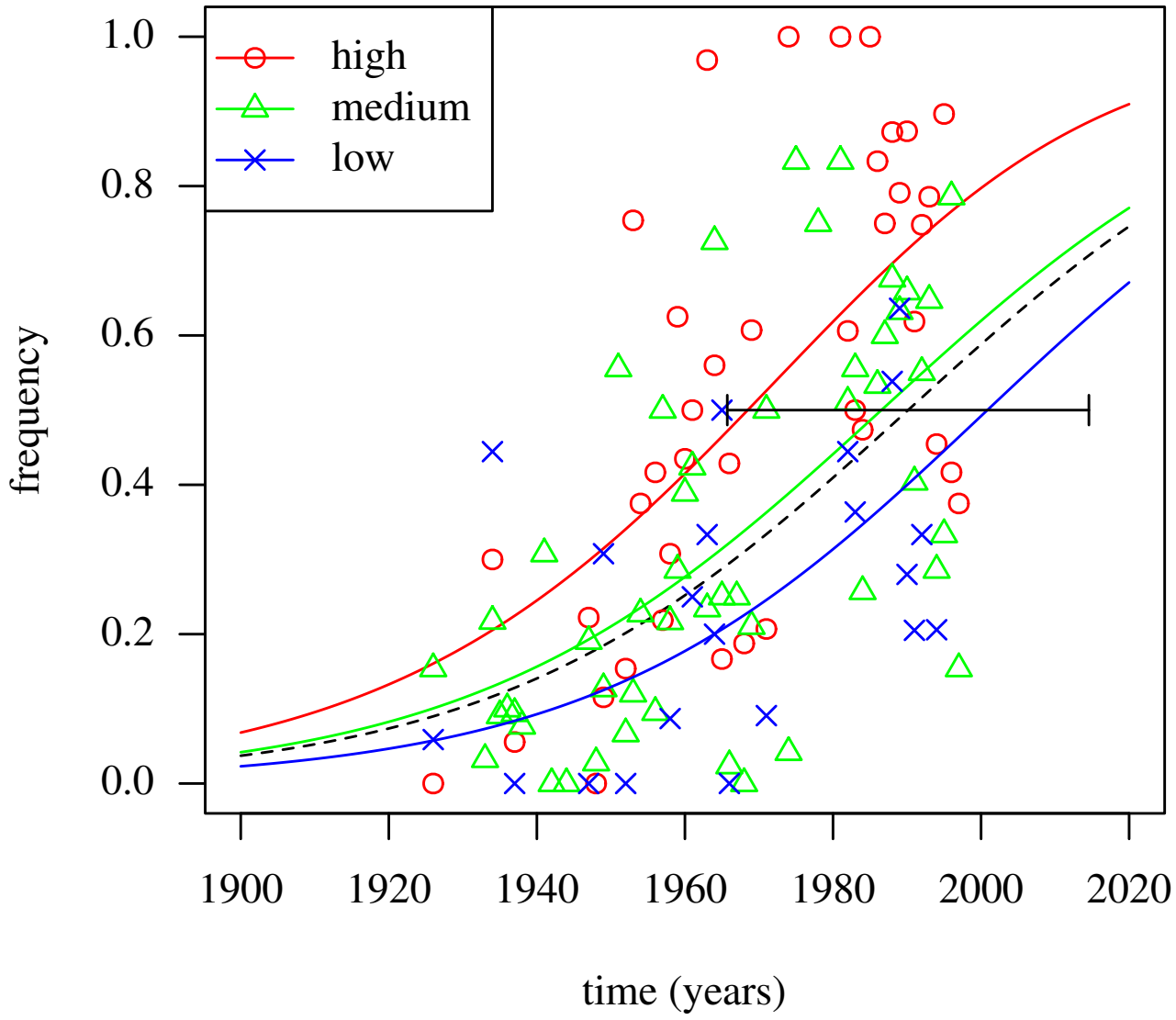
English Jespersen cycle (Wallage, 2013)



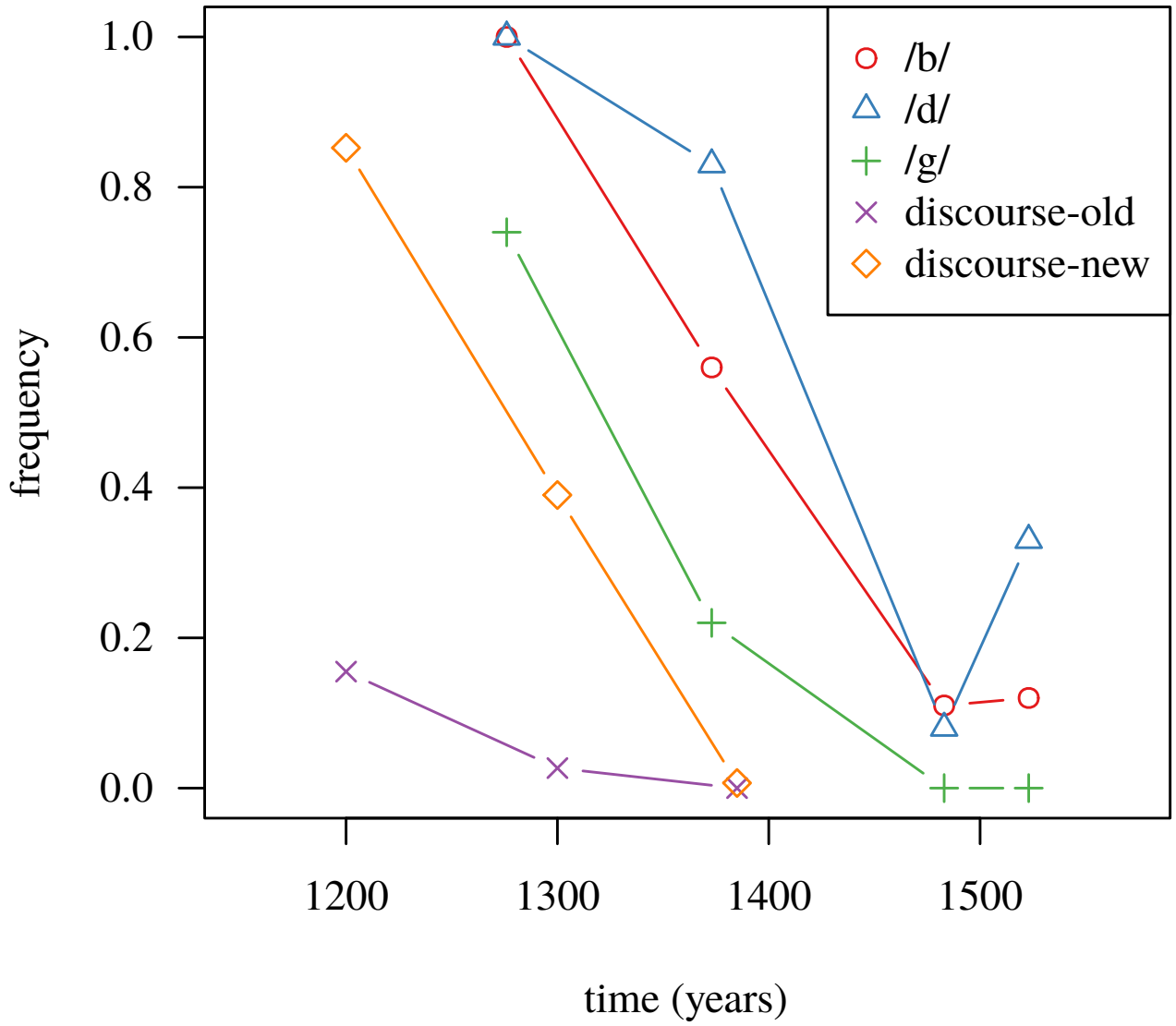
German final fortition (Fruehwald et al., 2009)



Manchester /t/-glottalling (Bermúdez-Otero et al., 2016)



Pseudo-CRE


1.0
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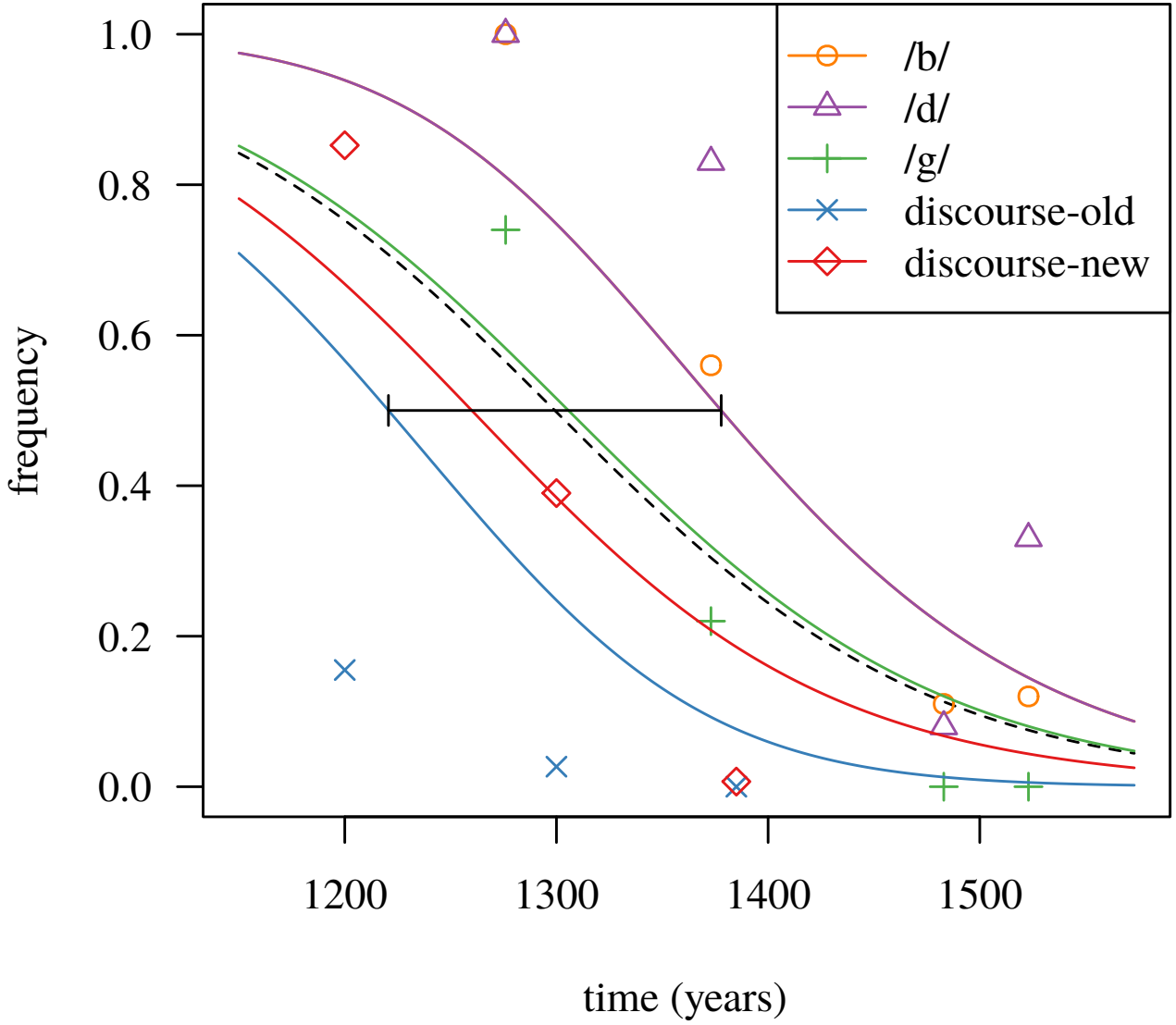
frequency
1200
1300
1400
1500
time (years)
/b/
/d/
/g/
discourse-old
discourse-new

Pseudo-CRE


1.0
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frequency
1200
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time (years)
/b/
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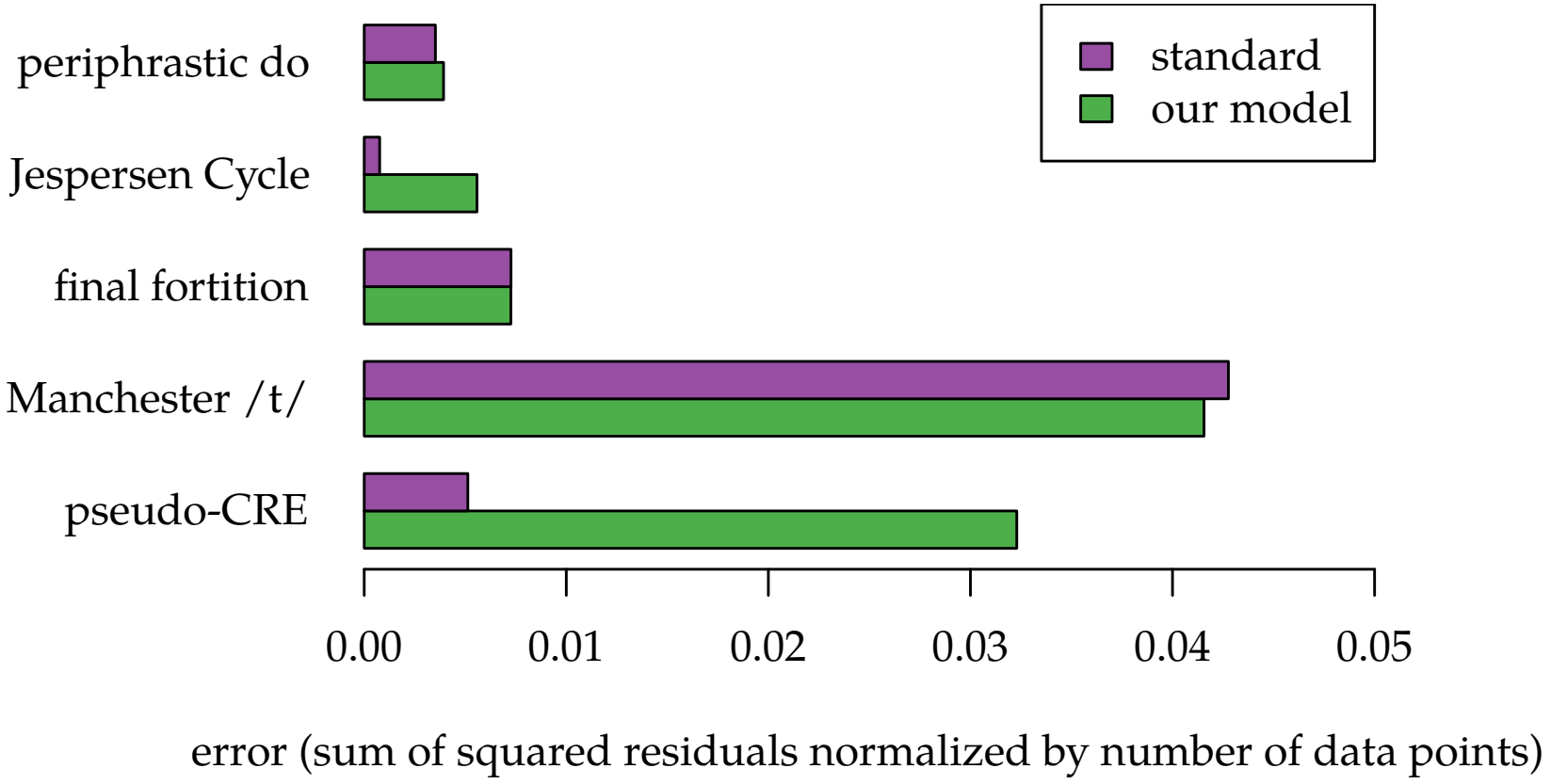
/g/
discourse-old
discourse-new

Fit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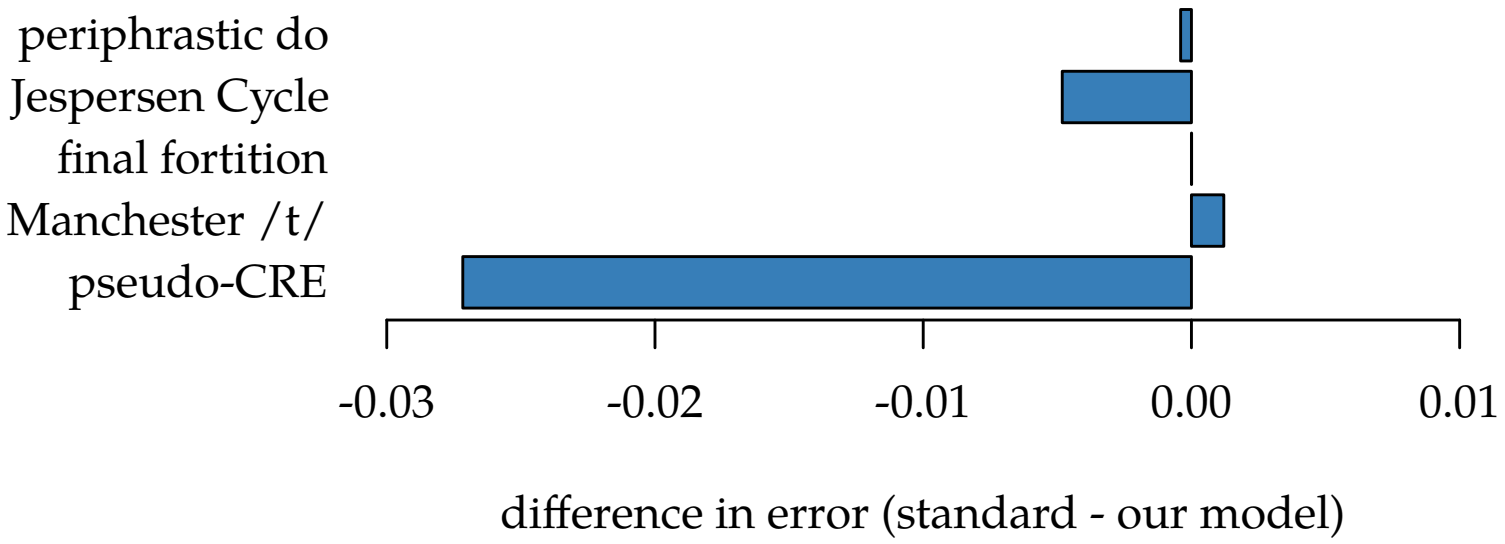
error (sum of squared residuals normalized by number of data points)

Fit (relative)

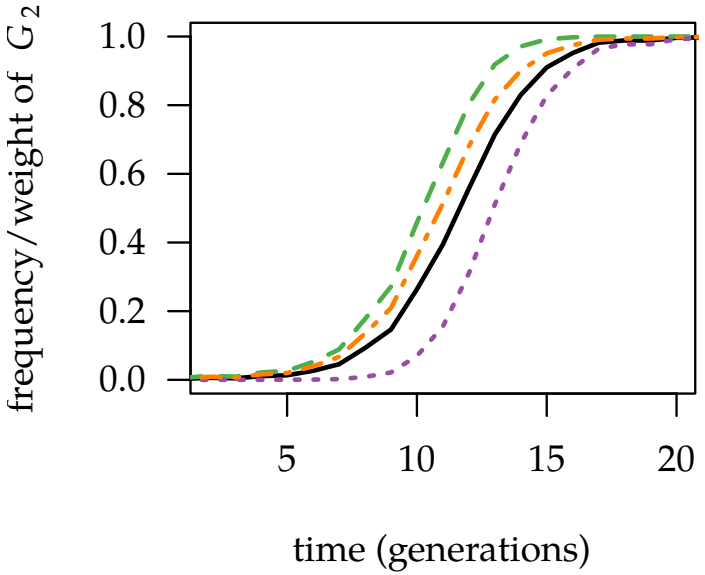


Part IV

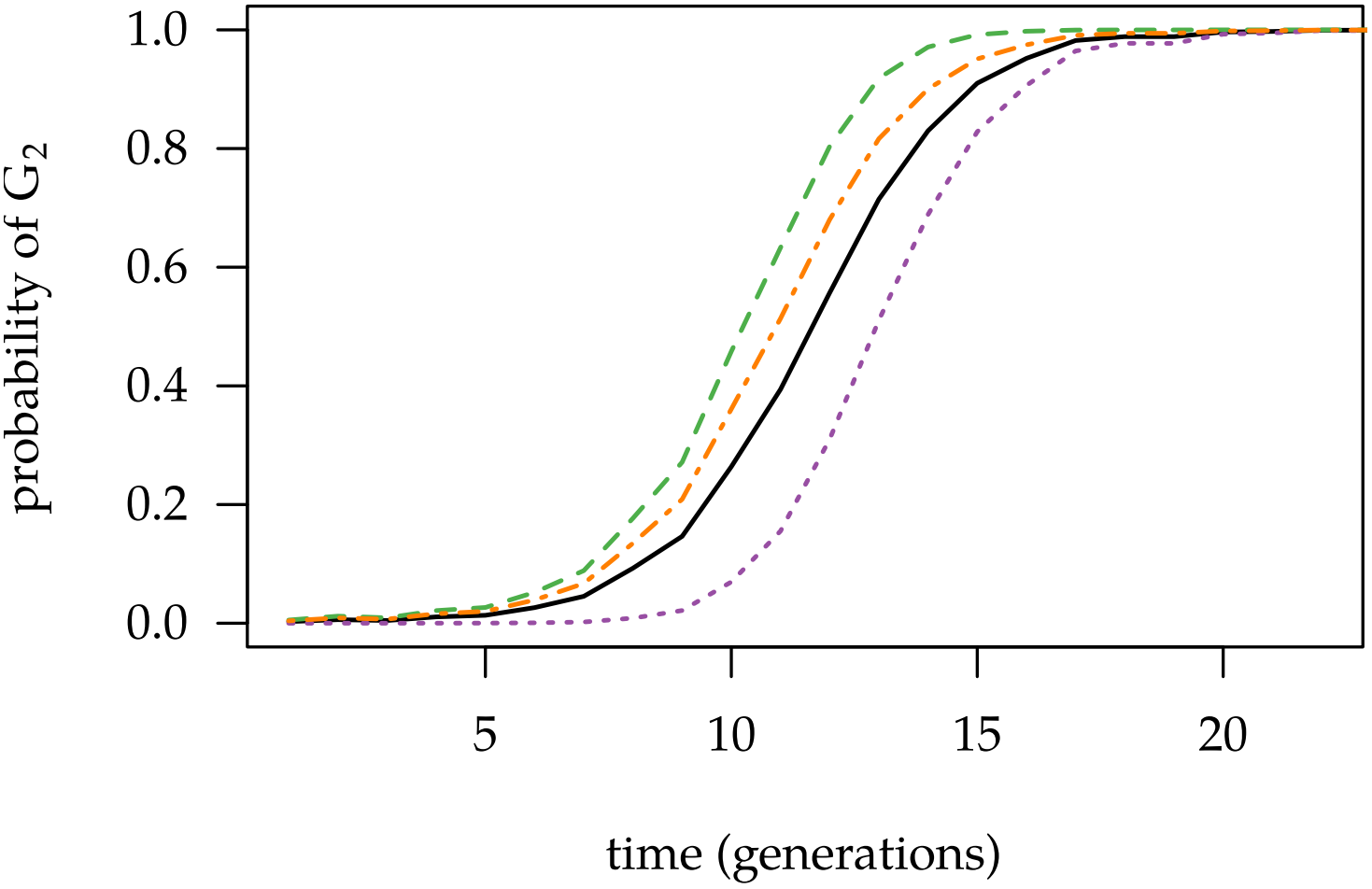
A note on the inconstancy of "constant" rates

A surprising(?) consequence

- When the Constant Rate Hypothesis is derived from first principles, slopes need not always be exactly equal across contexts
- Epistemologically, this implies that traditional methods may yield not only false positives (pseudo-CREs) but also **false negatives**



- For instance, a statistically significant time:context interaction in a logistic regression GLM need not refute the CRH, as we will now demonstrate


1.0
0.8
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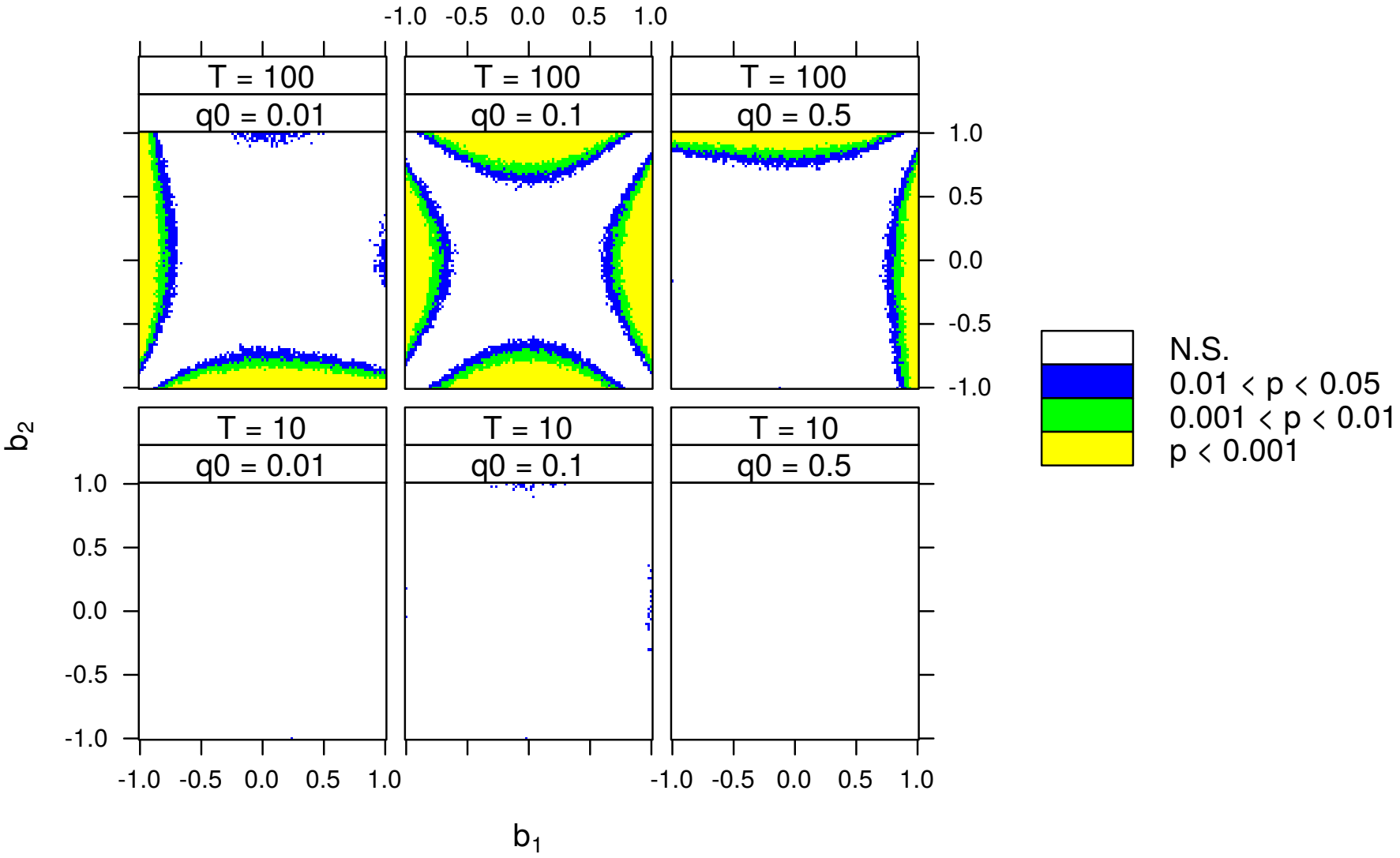
0.4
0.2
0.0
probability of G_2
5
10
15
20
time (generations)

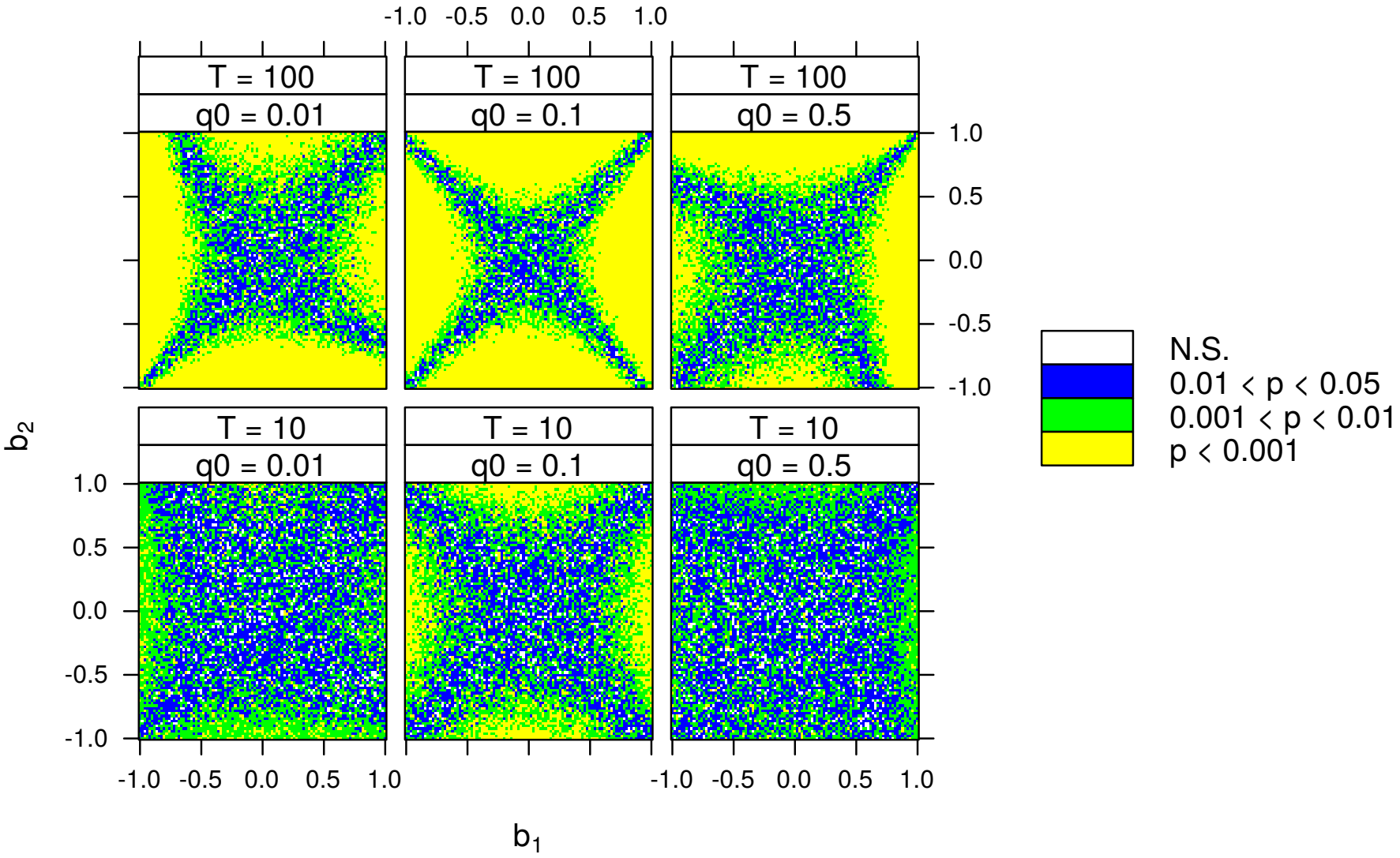
A surprising(?) consequence

- Let's generate artificial diachronic data from our model and track the evolution of $q_i(t)$ for two contexts $i = 1, 2$ by sampling T data points from each generation
  - Technically: each generation is a binomial experiment of T repetitions and probability of "success" $q(t)$, on which the production biases are applied
- This gives us usage frequencies in artificial "historical corpora" of varying data resolutions T
- We then run a logistic regression analysis on these time series and look for time:context interactions
- By systematically varying the bias magnitudes b_1 and b_2 we can control for their effect on the possibility of interactions

Interactions: means across 50 experiments



Interactions: maxima across 50 experiments

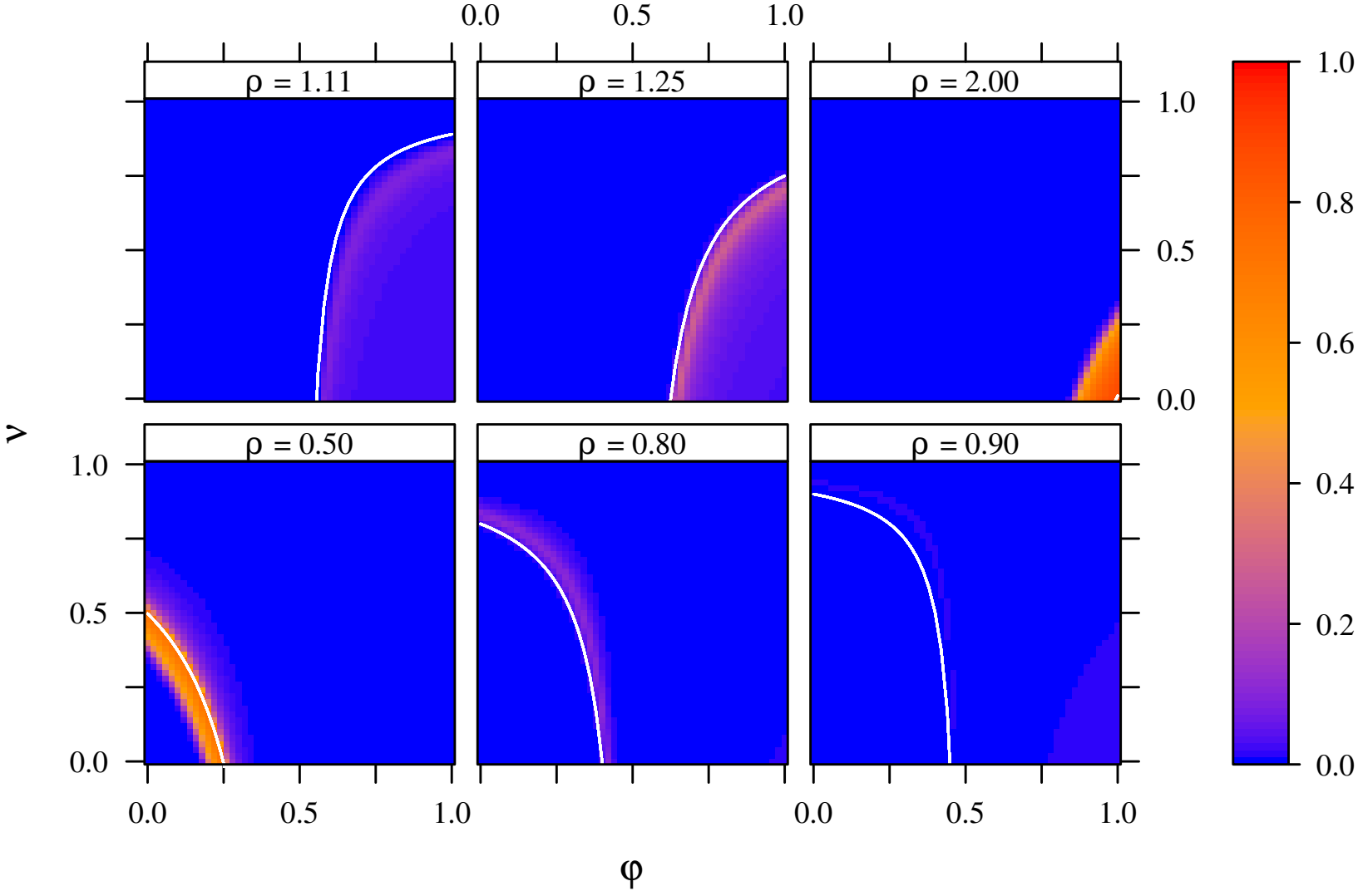


Interactions: lessons learned

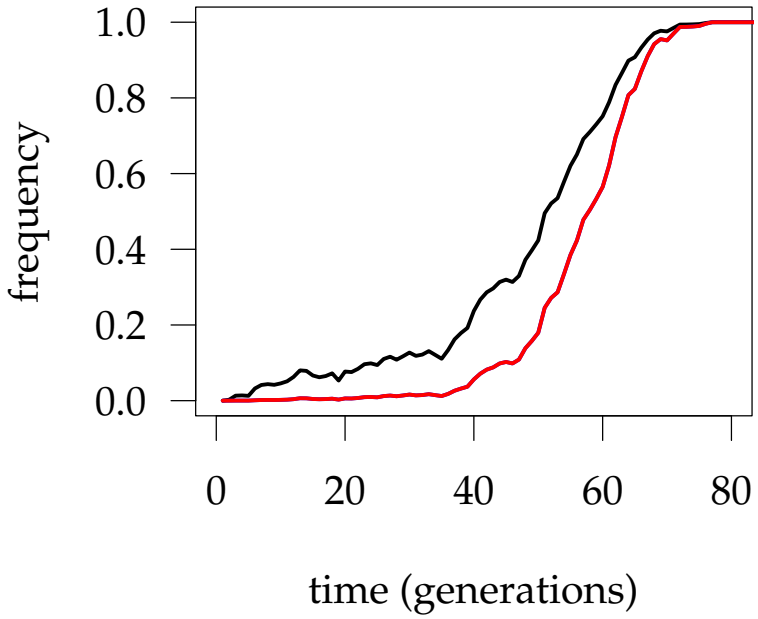
- Interactions are most likely to occur when $|b_1| \gg |b_2|$ or $|b_1| \ll |b_2|$, i.e. when the bias magnitudes for the two contexts differ greatly
- On average, one would not expect statistically significant time:context interactions when corpus is small
- However, even for small corpora, no region of the parameter space is **immune** to statistically significant time:context interactions
  - $\Rightarrow$ The presence of statistically significant time:context interactions in a logistic regression model of some diachronic data cannot be used to rule out the presence of a CRE, and therefore underlying grammatical unity of change, in those data

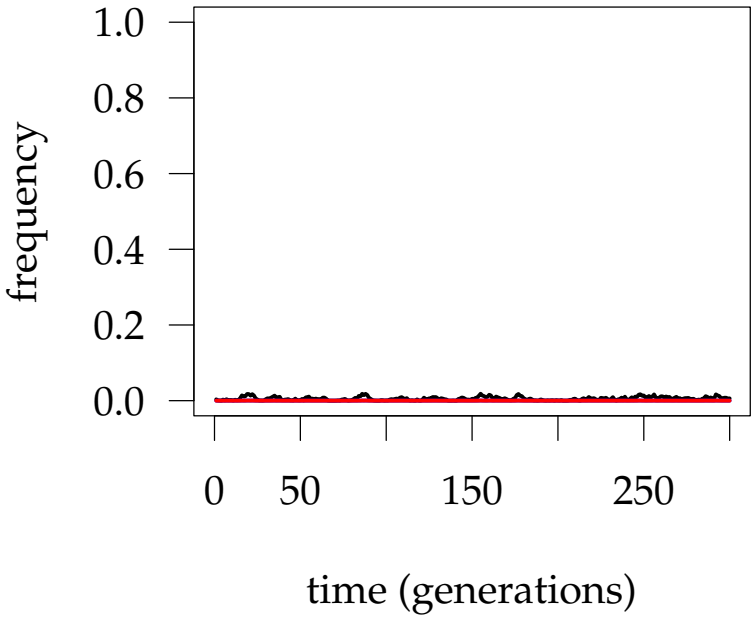
Part V

The non-logistic regime


ρ = 1.11
ρ = 1.25
ρ = 2.00
ρ = 0.50
ρ = 0.80
ρ = 0.90
ν
φ
0.0
0.5
1.0
0.2
0.4
0.6
0.8

Advantage vs. bias





- Sufficiently negative production biases can
  - slow down change and make it non-logistic (still S-shaped though!)
  - block change altogether

Advantage vs. bias

- So what, exactly, happens to the Fundamental Theorem?

Theorem (The Fundamental Theorem of Language Change)

Suppose learners are reliable (get a large number of input sentences and apply a reasonable learning rate). Then grammar G_2 overtakes grammar G_1 if, and only if, $\rho < 1$.

- Recall:
  - $\rho > 1$: the amount of data parsed by G_1 but not by G_2 is greater than the amount of data parsed by G_2 but not by G_1
  - $\rho < 1$: vice versa

Advantage vs. bias

Theorem (The Extended Fundamental Theorem of Language Change)

Assume reliable learners and write

$$B = \sum_{i=1}^{K} \lambda_i b_i \tag{8}$$

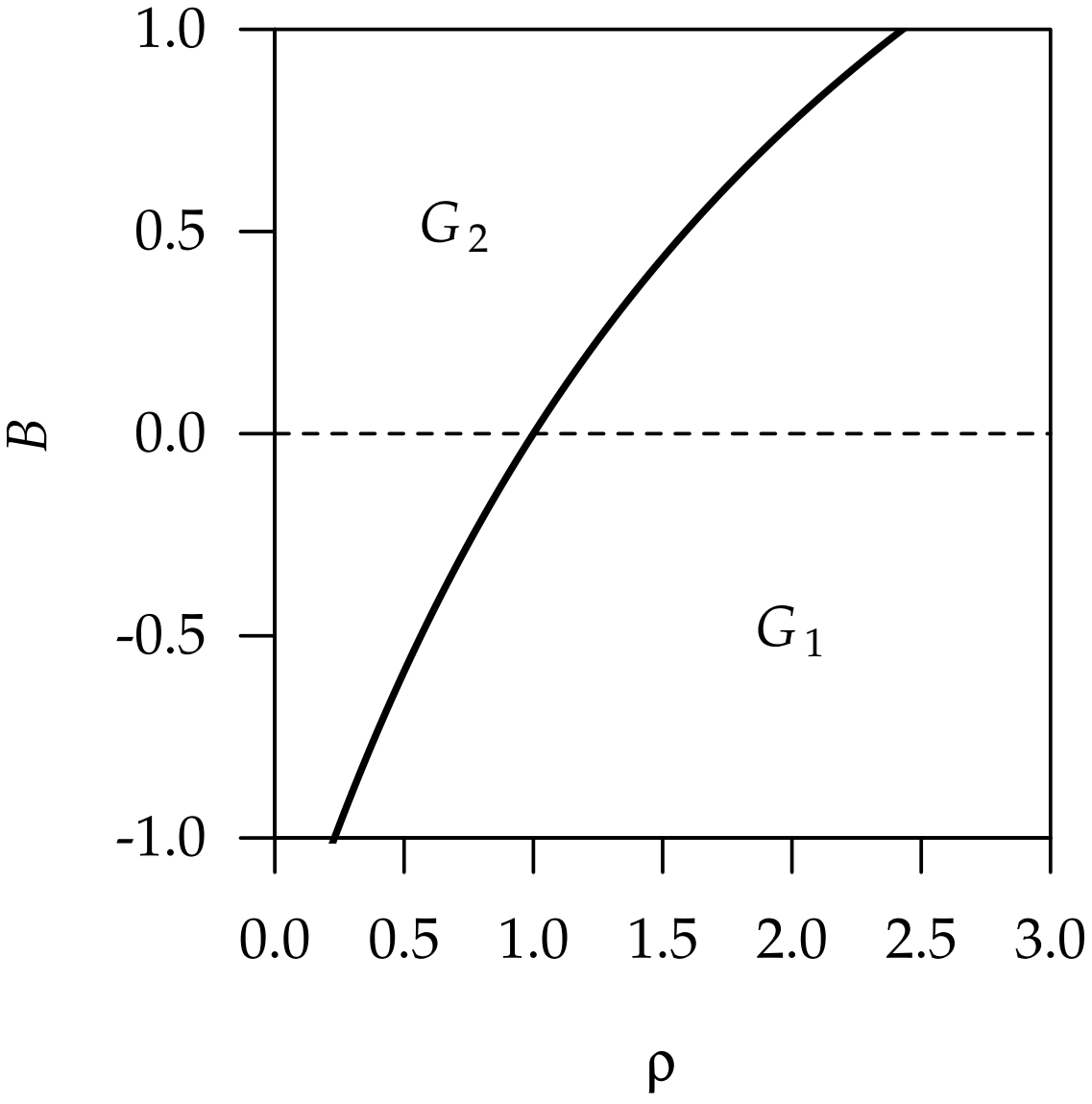
and

$$B_c = \frac{\rho - 1}{1 + q(0)(\rho - 1)}. \tag{9}$$

Then

1. $q(t) \to 1$ *as* $t \to \infty$, *if* $B > B_c$;
2. $q(t) = q(0)$ *for all* t, *if* $B = B_c$;
3. $q(t) \to 0$ *as* $t \to \infty$, *if* $B < B_c$.

In other words, G_2 *overtakes* G_1 *if, and only if,* $B > B_c$.


1.0
0.5
0.0
-0.5
-1.0
B
G_2
G_1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
ρ

Part VI

Conclusions

The three problems

- Problem 1: the traditional hypothesis allows arbitrarily large time separation between context curves.
  - Solution: Our model places an upper bound on this in a principled way.
- Problem 2: the traditional approach yields false positives.
  - (Partial) solution: Some of these are ruled out by the time separation theorem.
  - Observation: We shouldn't always expect the slopes to be the same.
- Problem 3: the traditional approach is based on unwarranted independence claims.
  - (Partial) solution: Move away from null hypothesis significance testing to models derived from first principles, and compare residual errors of fit instead.

Conclusions

- We have presented a **mechanism** for CREs, derived from simple ingredients:
  - The variational learner of Yang (2000)
  - Iterated learning (in an infinite, well-mixing population...)
  - Production biases with fixed weights
- A corollary of this is that an upper limit can be placed on the time separation of context curves (traditionally k).
- The model gives a good fit for a number of case studies.
- In the model, interactions with time are not necessarily evidence for underlying grammatical disunity.
- In some instances, production biases can outweigh the parsing 'advantage' of a grammar, so that it does not triumph in diachrony.

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